

Compression-bending behavior of thin-walled prestressed concrete tower with horizontal joint for wind turbines: Experimental study and calculation model

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ABSTRACT

This paper presents an experimental and analytical study on the compression-bending behavior of prestressed concrete towers, considering the effect of horizontal joints and prestress. Since the horizontal joint between the concrete tower and the foundation are not reinforced with continuous rebars, but are merely connected through prestressing force, there is a possibility of joint opening under the large bending moment. Five prestressed concrete tower specimens were tested under compression-bending loads. The key parameter in the experiment was the prestress ratio. The results showed that the increase of prestress can significantly constraint the opening of horizontal joints and improve the ultimate capacity of the tower, but the ductility and self-centering performance are reduced. Compared to the specimen with the lowest prestress ratio 0.11, the specimen with the highest prestress ratio 0.44 showed an increase of 84.56 % on the ultimate capacity, but showed a reduction of 79.84 % on the ductility. The accuracy of two commonly used design methods were evaluated based on test results. The relative errors were 48.6 % and 19.3 %, respectively, which showed low accuracy and applicability. Finally, based on the equivalent plastic hinge model, an analytical study was conducted. A theoretical model of the load-displacement curve was presented and the theoretical results were compared with the test results, showing a relative error of 4.2 %. The result indicated that the theoretical model can effectively predict the compression-bending capacity. In general, the revealed compression-bending mechanism and the proposed theoretical model can provide a guidance for the engineering design of prestressed concrete towers.

1. Introduction

The steel-concrete hybrid tower consisting of the lower prestressed concrete tower and the upper steel tower (Fig. 1), is widely used to support large-power wind turbines, due to the high stiffness, large bearing capacity, and low cost. The advantage of bearing capacity is primarily provided by the lower prestressed concrete tower. The prestressed concrete tower is a prefabricated structure, and it is connected to the foundation through horizontal joints (Fig. 1), which are generally assembled by grout or adhesive. However, no vertical reinforcement of the tower continuously passes through the joint, resulting in very limited bond performance of the joint [1]. Under bending moments, the connection of horizontal joint is prone to failure and then the tower can hardly bear any load. Thus, the connection of horizontal joint is usually

enhanced by applying prestressing force in the engineering [2], as shown in Fig. 1. Prestressing force is typically applied by post-tensioned unbonded external tendons, which pass through the hollow interior of the tower to the foundation. One end of the tendons is anchored at the top of the concrete tower, and the other end is anchored at the foundation. This ensures a strong connection between the tower and the foundation, enhancing the structural integrity of the tower [3]. Thus, the effect of prestress on the performance is significant. Additionally, prestressed concrete towers are typically designed with a circular cross-section with a high hollow ratio χ , which is ranged from 0.75 to 0.95, making the towers be classified as thin-walled concrete structures.

Aerodynamic wind load is the most primary load that prestressed concrete towers withstand. The aerodynamic wind load can be simplified as the aerodynamic thrust F acting on the top nacelle, which

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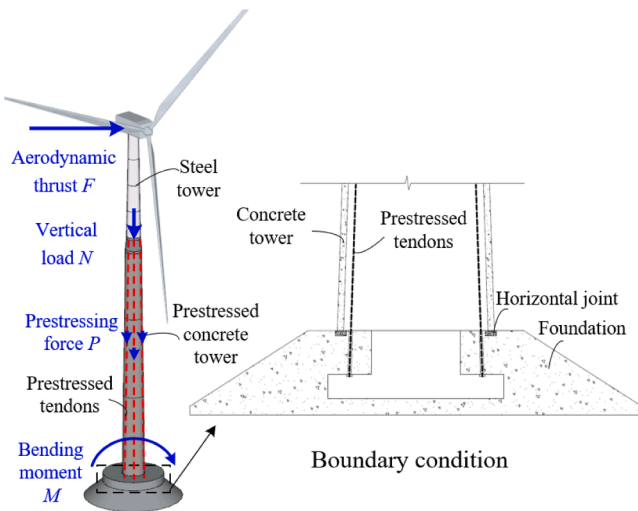


Fig. 1. Schematic of the hybrid tower and prestressed concrete tower.

generates a bending moment M on the lower concrete tower. Since the applied location of the aerodynamic thrust is usually far from the lower prestressed concrete tower, the shear span ratio is high and the shear effect can be ignored. Besides, the tendons apply a significant prestressing force P to the tower, and the self-weights of the wind turbine and the tower generate the vertical load N . These two forces form a compression on the tower, but the vertical load N is typically much slighter than the prestressing force P . In summary, the prestressed concrete tower primarily subjects to the compression-bending load (Fig. 1), which is also the main load for most wind turbine towers [4]. In recent years, the loads have significantly increased with the rapid development of wind turbines. When the bending moment becomes too large, the prestress may no longer keep the tower section in a fully compressive state, causing the horizontal joint to open. This results in that a part of the joint remains compression while the other part forms a gap. In this case, the joint interface is no longer a complete circular cross-section, significantly increasing the geometric nonlinearity of the tower. The mechanical behavior becomes more complex, but the tower can still continue to bear bending moments with the joint partially opened. Therefore, it is essential to consider the effect of the joint opening when analyzing the mechanical behavior of the prestressed concrete tower, which should be investigated in detail.

Most previous studies tended to focus on the conceptual and scheme design of prestressed concrete towers [5–7], and the analyzes emphasized overall structural responses such as frequency and vibration behavior [8–11]. Additionally, optimization design of the tower based on indexes such as strength, stability, and dynamic response have also been focused [12–15]. Huang et al. [16] introduced the methods for calculating the bearing capacity of prestressed concrete towers, but only the elastic strength was checked, which could not accurately assess the load-bearing performance. Haar [17] Grünberg [18] provided a method for calculating the bending capacity based on European code [19], but neither of them considered the effect of joint opening. Considering the lack of existing engineering design methods, several experimental studies on prestressed concrete towers can be found in recent years. Li et al. [20] conducted a lateral loading test on a hybrid tower specimen with a scale ratio of 1:10, but the horizontal joint was not considered in the specimen. No destructive loading was conducted, making it impossible to analyze the ultimate mechanical behavior of the tower. Ma et al. [21] conducted experimental investigation on a 5 MW prestressed concrete tower, testing the mechanical behavior and the dynamic performance. But the joint was not considered and no destructive loading was carried out neither. Zhang et al. [22] conducted a large-scale model test with a scale ratio of 1:3, considering details such as horizontal joints

and vertical joints. However, only the verification loading based on scaled loads was performed, and the ultimate behavior was not analyzed. It can be found that most previous studies ignored the effect of the horizontal joint, which failed to account for the nonlinear effects after the joint opening. A few studies that considered the effects of the joints only performed analyzes under scaled actual loads nor than destructive loading, making it impossible to study the ultimate behavior of the tower.

The lack of study on prestressed concrete towers highlights the need of referencing findings from other similar structures. The precast bridge pier also includes horizontal joints and post-tensioned unbonded prestress tendons. The use of unbonded post-tensioning was intended to achieve self-centering performance and to reduce energy dissipation, thereby enhancing seismic performance. Although the design intentions for using post-tensioned prestress are different, both the prestressed concrete tower and the precast bridge pier exhibit the behavior that joint opens under significant bending moments, while the joint closes upon unloading. This nonlinear behavior has a similarly impact on structural performances. Therefore, the research findings on post-tensioned unbonded precast bridge piers can provide valuable insights for this study. Mander and Cheng [23] conducted experimental studies on the seismic behavior of post-tensioned unbonded precast bridge piers, finding that the bending capacity mainly depended on the prestressing force and the vertical load. Lin and Mo [24] conducted cyclic tests on hollow precast bridge piers, moving the joint from the base to the midsection of the specimen, creating a joint between two precast segments. The analysis indicated that the plastic behavior of the pier could not be affected by the joint since it was away from the critical region at the base, highlighting the critical role of the bottom joint. Hewes et al. [25] conducted cyclic tests on circular bridge piers, considering the effects of prestress and shear span ratios. The results showed that the damage of the bridge pier was slight and primarily concentrated in the area near the bottom joint. Ou et al. [26] introduced the concept of the decompression zone in precast bridge piers, where the compressive stress was completely offset and the joints were opened. Bu et al. [27,28] conducted cyclic tests on multi-segment precast bridge piers, further demonstrating that even when the pier was composed of multiple segments and joints, damage still concentrated at the bottom joint, while damage at the upper joints is minor. Based on the test results, the analytical method for establishing the relationship between lateral force and displacement was derived [29,30]. Existing research on precast bridge piers is quite abundant, but there are still differences in structural characteristics compared to prestressed concrete towers. Prestressed concrete towers generally feature hollow ratios greater than 0.75 [31], resulting in a thin-walled circular section. Under bending moments, towers are prone to compressive failure. Additionally, the prestressed tendons in concrete towers are uniformly distributed along the hollow interior of the circular section and close to the wall. The deformation characteristic differs significantly from that of centrally located tendons in precast bridge piers. Furthermore, the prestressing force is considerably greater than the structural weight, which will have a different impact on the geometric nonlinearity of the joints.

It can be found from above studies that the compression-bending behavior of prestressed concrete towers considering the effect of horizontal joint is still unclear. Besides, there is a lack of destructive experimental studies on prestressed concrete towers, and the analytical methods for the ultimate capacity remain rare. To address above gaps, an experimental study and analytical study were conducted on the compression-bending behavior of prestressed concrete towers in this paper. The load-bearing performance and hysteretic behavior of prestressed concrete towers with horizontal joints were analyzed through destructive tests. The accuracy and applicability of existing design methods were evaluated based on the test results. A theoretical analysis model was developed by combining the equivalent plastic hinge model and ultimate equilibrium theory. This model can be used to predict the load-displacement curve of prestressed concrete towers, and the

theoretical results were validated by measured results. This study can provide a guidance for the engineering design of prestressed concrete towers.

2. Experimental programme

2.1. Specimens design

Five prestressed concrete tower specimens were designed under compression and bending load. All specimens were kept at the same size to directly compare the performance of prestressed concrete towers with different prestress ratios and load methods. The scaled specimens were designed considering the geometric and mechanical characteristics of actual prestressed concrete towers, as well as the constraints of specimen fabrication and loading conditions in the lab, as shown in Fig. 2. The sectional dimensions were scaled down by a ratio of 1:8, based on the concrete tower section of a H140m-2.0 MW-120 m hybrid tower. The diameter and thickness of specimens were 610 mm and 55 mm, respectively. The shear span ratios of prestressed concrete towers are usually large, resulting that the bending moment is the main load and the shear force can be neglected. Therefore, the focus in designing specimen height was to avoid the shear effects and ensure the application of combined compression and bending loads. According to reference [32], the shear effects can be disregarded when the height-diameter ratio (shear span ratio) exceeds four. Consequently, the specimen height was designed at 2600 mm, including a 2400 mm standard tower section and a 200 mm transition section, which can effectively minimize the shear effect. The taper of concrete towers in practical engineering is minimal, leading to an insignificant impact on tower's load-bearing performance. On the other hand, many towers adopt straight form without taper. Consequently, the specimens were designed as straight tower for the ease of specimen processing and fabrication.

The transition section at the top of the specimen, was designed to anchor the prestressed tendons and connect to the test setup. To simulate the grouted horizontal joint between the tower and the foundation, a solid concrete base with 300 mm height and 1200 mm diameter was

designed. A circular groove was created at the top of the base, which was filled with high-strength grout. As shown in Fig. 2, the interface between the specimen and the concrete base was only bonded by the grout, without vertical reinforcement passing through. The connection between the specimen and the concrete base was primarily achieved through the prestressing force. The higher the level of prestress, the tighter the connection between the specimen and the base. The higher-strength concrete and grouting materials were used for the base and horizontal joint, respectively, to prevent that they crush before specimens. Specific parameters for each material are detailed in Table 2.

The reinforcement of the specimen was designed based on the configuration of actual tower, utilizing double layers of vertical reinforcement and circumferential reinforcement. The vertical reinforcement ratio was set at 1.00 %, while the circumferential reinforcement ratio was set at 0.61 %. Considering that the transition section needed to withstand localized pressure applied by prestress tendons, the reinforcement in the transition section was designed denser to prevent failure before that of the tower section. The schematic diagram of specimen reinforcement is shown in Fig. 3. The diameter of reinforcement bars was 4 mm, and the material properties can be found in Table 3.

Eight external unbonded prestressed tendons were uniformly distributed around the cross-section. The top ends of the prestressed tendons were anchored to the top surface of the transition section, passing through the hollow interior of the tower section, while the bottom ends were anchored to the bottom surface of the concrete base, as shown in Fig. 2. Single steel strand with diameter of 15.2 mm was adopted for each tendon.

The parameters of specimens were prestress ratios (n) and loading methods. The prestress level is quantified using the prestress ratio (n), calculated as $n = P/A_c f_c'$, where P is the total prestressing force applied to the specimen, A_c is the sectional area, and f_c' is the axial compressive strength of the concrete. The prestress ratios were ranging from 0.11 to 0.44 to compare the effects of different prestress levels. The loading methods were categorized into monotonic loading and cyclic loading. The detailed parameters are listed in Table 1.

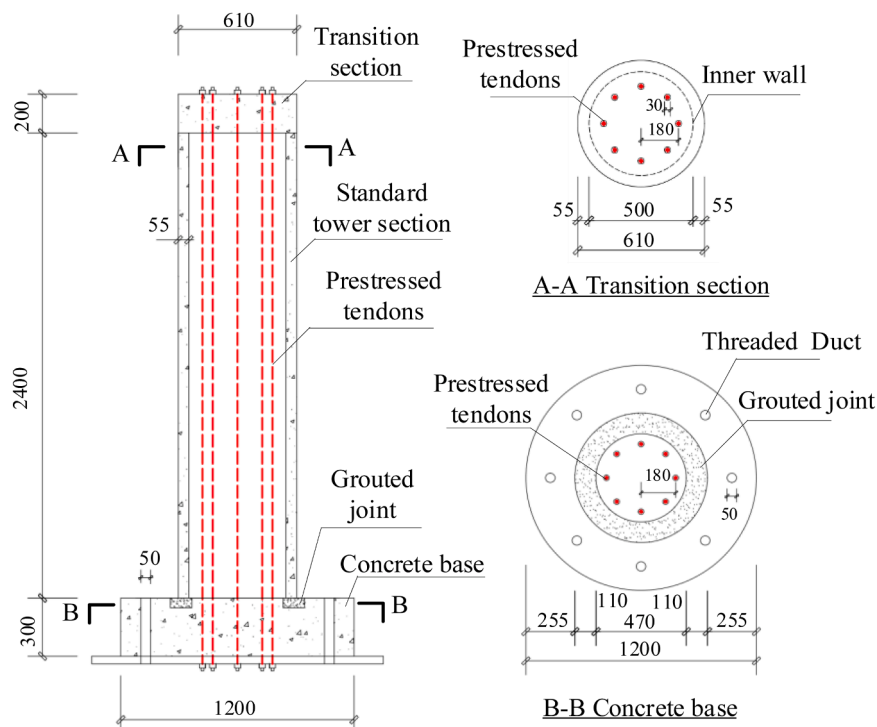


Fig. 2. Schematic diagram of test specimens.

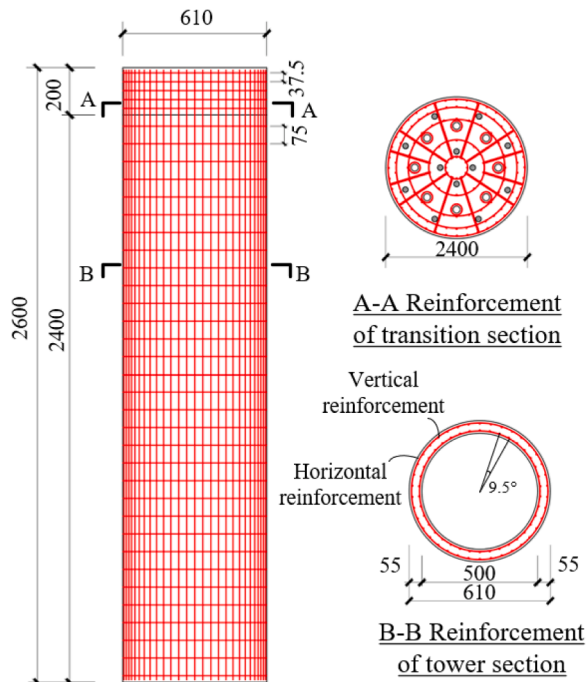


Fig. 3. Specimen reinforcement.

2.2. Specimens processing

The reinforcement bars were cut and positioned according to the designed dimensions. Aluminum formwork was manufactured according to the specimen dimension, ensuring a smooth surface to achieve

uniform concrete quality. Due to the thin walls of specimens, fine aggregate concrete with smaller particle sizes was used for pouring. Denser reinforcement bars were assembled for concrete bases. Additional spiral reinforcement was tied around the prestress ducts and threaded ducts to prevent localized compression damage. Furthermore, more vertical and circumferential reinforcements were tied below the grouting groove to prevent punching shear damage. PVC pipes were embedded at the predetermined duct locations. The formwork was then supported, and concrete was poured. Following a natural curing period of 28 days, the specimens and concrete bases were completed. Fig. 4

After that, the grouting of the horizontal joint was conducted. To ensure a level surface at the bottom of the specimen after grouting, three leveling pads were poured into the groove, evenly distributed in a circular arrangement, as shown in Fig. 5(a). The leveling pads were poured using the same grouting material. The specimens were positioned on the pads, ensuring precise alignment to guarantee that specimens were centered on the base. After that, self-compacting high-strength grout without requiring vibration was poured into the groove, as shown in Fig. 5(b). The excess grout on the surface was removed, and the joint construction was completed after natural curing.

2.3. Material properties

Six cubic samples were reserved for specimen, base, and grout to determine their compressive strength, respectively. The cubic samples were fabricated in accordance with GB/T 50081-2002 [33]. The average compressive strength for the concrete of specimen, concrete of base, and grout are presented in Table 2. Additionally, three prismatic samples were fabricated to test their elastic modulus. The fabrication and testing methods were also based on the guidelines of GB/T 50081-2002. The average values of the elastic modulus are presented in Table 2.

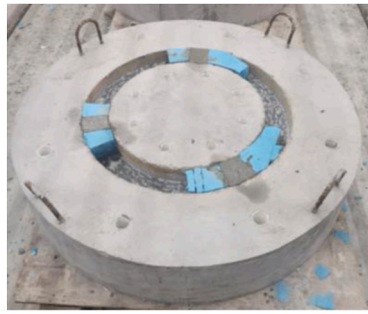
The material properties of the reinforcement bars and steel strands were tested according to the guidelines of GB/T 228.1-2010 [34]. Due to

Table 1
Specimen parameters.

Specimen label	Loading method	Prestress ratio n	Designed prestressing force (kN)	Actual prestressing force P (kN)	Average initial prestress (MPa)
CB-0.11-c	Cyclic loading	0.11	417	460.4	411.1
CB-0.22-c	Cyclic loading	0.22	834	848.6	757.6
CB-0.33-c	Cyclic loading	0.33	1251	1233.4	1101.3
CB-0.44-c	Cyclic loading	0.44	1668	1556.6	1389.9
CB-0.44-m	Monitonic loading	0.44	1668	1546.0	1380.4



Fig. 4. Reinforcement assembling and concrete forming.



(a) Leveling pads in the joint groove



(b) Joint grouting

Fig. 5. Grouting of horizontal joint.

Table 2
Material properties of concrete and grout.

Material type	Concrete of specimen	Concrete of base	High-strength grout
Cube compressive strength	47.27 MPa	69.81 MPa	83.52 MPa
Elastic modulus	32,687.67 MPa	37,528.25 MPa	39,767.30 MPa

the large dimension and concrete strength of the base, the reinforcement strength of the base had minimal impact on the test results. Therefore, the material property test of the reinforcement in the base was ignored. The measured average elastic modulus, yield strength, and ultimate strength for the reinforcement bars and steel strands can be found in Table 3.

2.4. Test setup

The test setup mainly consists of a loading beam, a cross hinge, an actuator, a steel cover, and a steel base, as shown in Fig. 6. The concrete base was tightly connected to the steel base using M46 high-strength bolts. Twelve M24 high-strength bolts were embedded in the top of the specimen (transition section), to connect the specimen to the steel cover. Additionally, the prestressed tendons were anchored to the steel cover to apply prestressing force to the specimen. The top surface of the steel cover was connected to the bottom of the loading beam using high-strength bolts. One end of the actuator was connected to the reaction wall, while the other end was connected to the loading beam through a cross hinge, allowing for the application of lateral force. Several holes were set on the loading beam to connect with the cross hinge, allowing the actuator to be connected to the central of the loading beam. As described in Section 2.1, the height-diameter ratio of the specimen is large, allowing shear effects to be neglected. Therefore, the lateral force F applied by the actuator only introduced a bending moment M to the specimen. With the prestressing force P applied by the steel strands, the combined compression-bending load could be achieved, as shown in Fig. 6(c).

Table 3
Material properties of steel.

Material type	Yield strength f_y (Mpa)	Ultimate strength f_u (Mpa)	Elastic modulus E_s (Mpa)
Reinforcement bars	523.18	580.65	199,325.2
Steel strand	1804.86	1996.43	204,021.3

2.5. Measurements and loading scheme

The main measurements include the lateral force F applied by the actuator, the prestressing forces $P_1 \sim P_8$ applied by tendons and the horizontal displacement Δ . The lateral force and prestressing forces were measured by force sensors, while the horizontal displacement was measured with a linear variable differential transformer (LVDT). The measurement arrangement is illustrated in Fig. 7.

After the prestress was applied, loading was conducted in two forms: cyclic loading and monotonic loading. A pre-loading step was performed before formal loading, to avoid virtual displacement and to ensure that all instruments were functioning correctly. For the cyclic loading, the loading scheme was controlled by the displacement increment. Each displacement increment was set to ± 3 mm. After reaching the corresponding displacement of each level, a holding time of 2 min was implemented to facilitate data collection and phenomena observation. For the monotonic loading, loading was performed only in the positive direction with a constant speed 0.5 mm/min. Loading was stopped when the load decreased below 85 % of the peak load or when excessive deformations occurred that made further loading inadvisable.

3. Experimental results and discussion

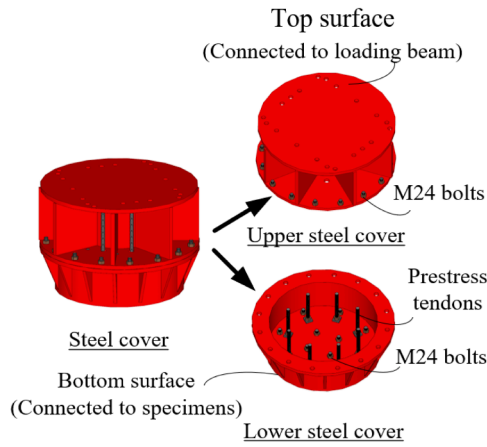
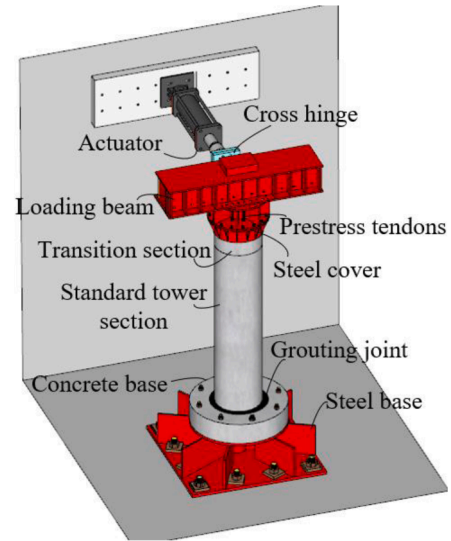
3.1. Test phenomenon and failure mode

All specimens exhibited the phenomenon of joint opening and crush of the concrete. However, the variations in prestress ratios led to differences in the specific phenomenon observed among the specimens. The failure modes of specimens are shown in Fig. 8. In the initial loading stage, transverse cracks appeared at the bottom of the tensile side of the specimen. Subsequently, the tensile side was decompressed and the joint was opened, while vertical cracks began to form at the bottom of the compressive side. The opened joint would close during reverse loading. The transverse cracks of the tensile side exhibited later on specimens with higher prestress ratios, while vertical cracks of the compressive side appeared earlier, with longer and wider cracks. For the specimen with the highest prestress ratio (0.44), long vertical cracks appeared after the prestress was applied, which extended during loading. As loading progressed, the joint opening on the tensile side increased, and the concrete spalling occurred on the compressive side. The joint repeatedly opened and closed throughout the cyclic loading process. In specimens with lower prestress ratios, the phenomenon of opening and closing of the joints were more pronounced.

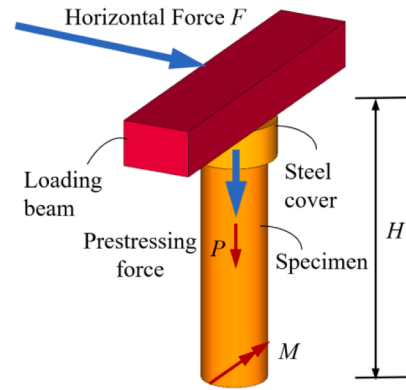
The concrete near the vertical cracks at the bottom of the compressive side tended to bulge outward when the load became excessive. Shortly after, the concrete rapidly crushed and spalled, causing the vertical reinforcement to buckle and the transverse reinforcement to bulge outward. The load then dropped rapidly. Specimens with higher prestress ratios exhibited sudden concrete crush with a larger crushing area, and the joint opening on the tensile side was relatively small.



(a) Overall diagram

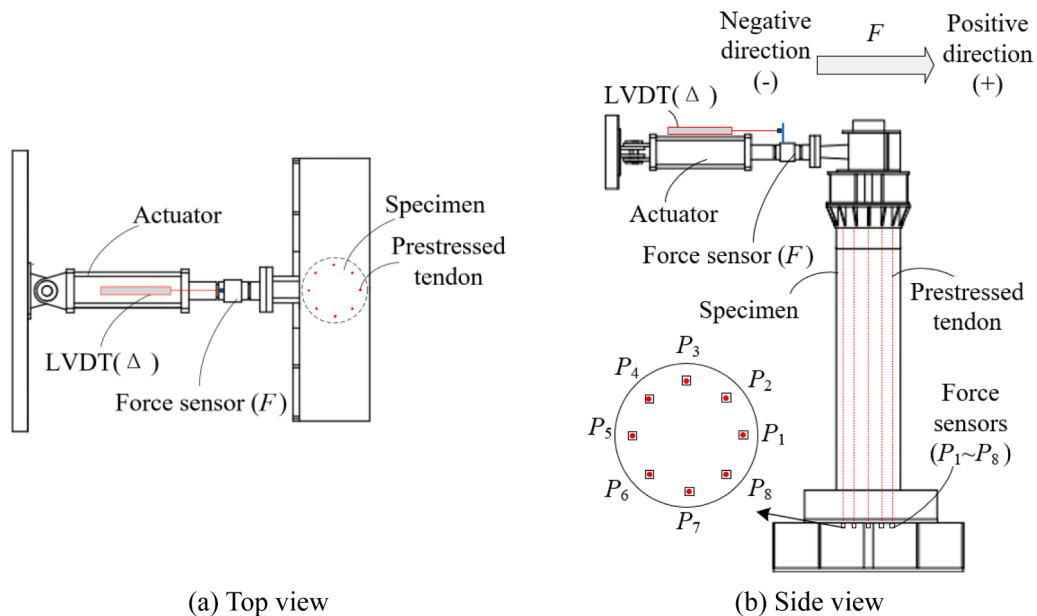


(b) Steel cover



(c) Loading mode

Fig. 6. Test setup.



(a) Top view

(b) Side view

Fig. 7. Measurement scheme.

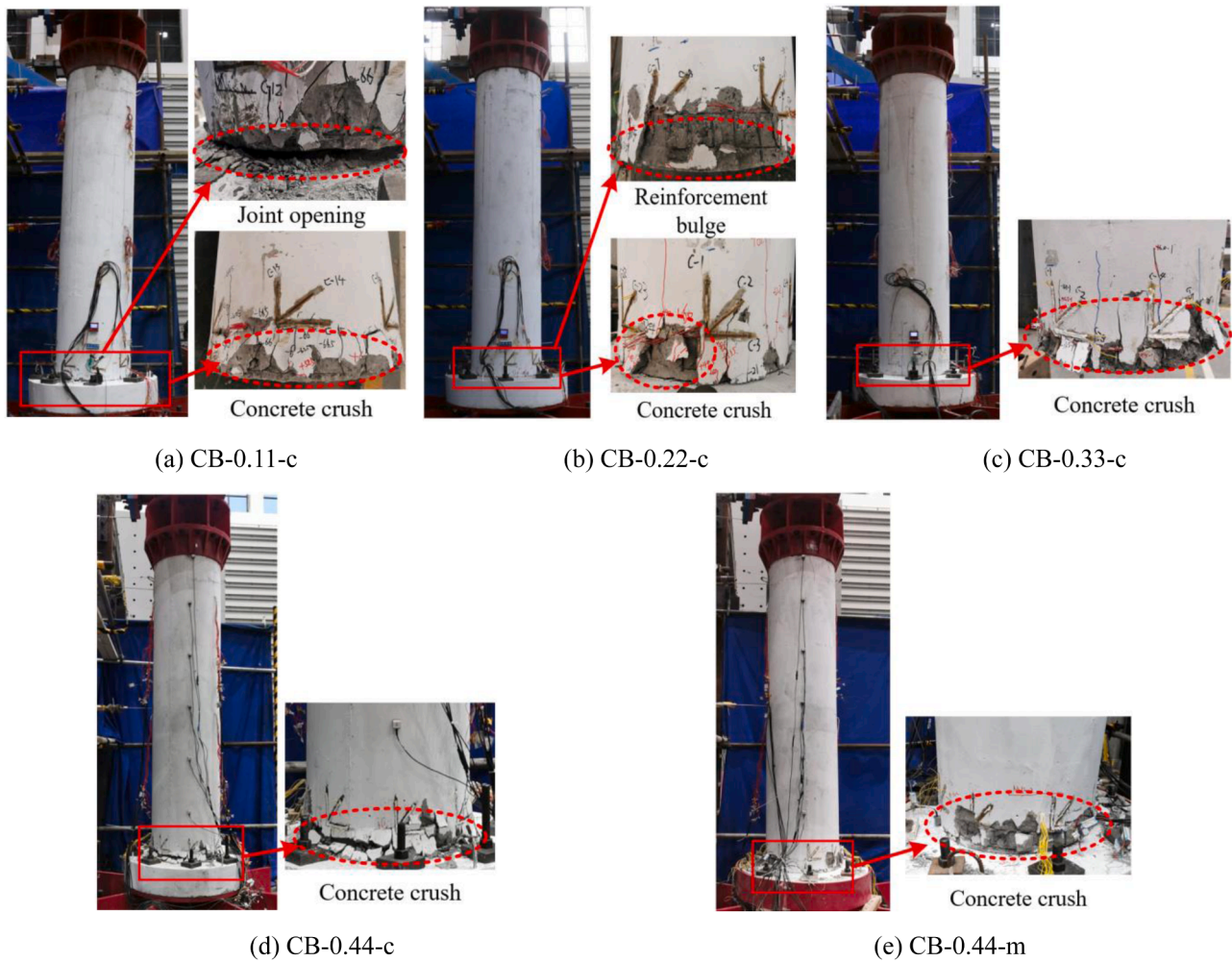


Fig. 8. Failure modes.

Specimens with lower prestress ratios showed gradual crushing and spalling of the concrete at the bottom section, with the spalled concrete being relatively small. The load was still in the ascending stage when the concrete on the compressive side began to spall. The crushing process was slower, and the joint opening on the tensile side was more significant. After failure, no transverse crack appeared in the tensile region of the specimen, and the rest regions of specimens remained largely intact. Additionally, no slippage was observed at the joint. Therefore, the typical failure mode of the prestressed concrete tower specimen can be summarized as follows: the concrete in the bottom region exhibited significant compressive damage and subsequently crushed, while the horizontal joint opened.

3.2. Hysteretic curves

The hysteretic curves of specimens are shown in Fig. 9. The hysteretic loops exhibited a linear trend with high stiffness in the early stages. Combined with the experimental observations, this indicates that specimens were in the elastic phase where the joints had not yet opened. After the joint opened, the rate of load increase significantly slowed down, and the curve exhibited yielding characteristics. However, the loading and unloading processes remained linear with similar stiffness, and the unloading path only deviated slightly from the corresponding loading path. After unloading, the residual displacement and the area of the hysteretic loops were small, and the hysteretic curves showed significant pinching. The characteristics of the hysteretic curves indicate that the joints repeatedly opened and closed under cyclic bending

moments. After joints closed, specimens exhibited small plastic deformation, returning to the state before joints opened. The specimens demonstrated significant self-centering behavior, relying on the restoring effect of the prestress. By comparing the hysteretic curves of different specimens, it can be initially concluded that the specimens with smaller prestress ratios exhibited more pronounced self-centering behavior, and showed smaller loop areas. After reaching the peak load, specimens with small prestress ratios progressively dissipated energy and the load gradually decreased during this process. In contrast, specimens with larger prestress ratios showed little self-centering performance, with noticeable energy dissipation even before reaching the peak load. After reaching the peak load, the load dropped abruptly, leading to rapid failure without a ductile decrease phase.

3.3. Skeleton curves

The skeleton curve was obtained by selecting the maximum displacement points from each hysteresis loop. The skeleton curves of the specimens were compared in Fig. 10, and the key characteristic points of each specimen were presented in Table 4. The yield points of the specimens were determined according to [35]. The peak points were the point where the peak loads occurred, and the ultimate points were the points where the loads dropped to 85 %. For specimens with a rapid load drop, the characteristic point before concrete crush was taken as the ultimate point. It was observed that the skeleton curves showed S-shape, and could be divided into two stages based on stiffness change. The initial stiffness of different specimens was basically the same,

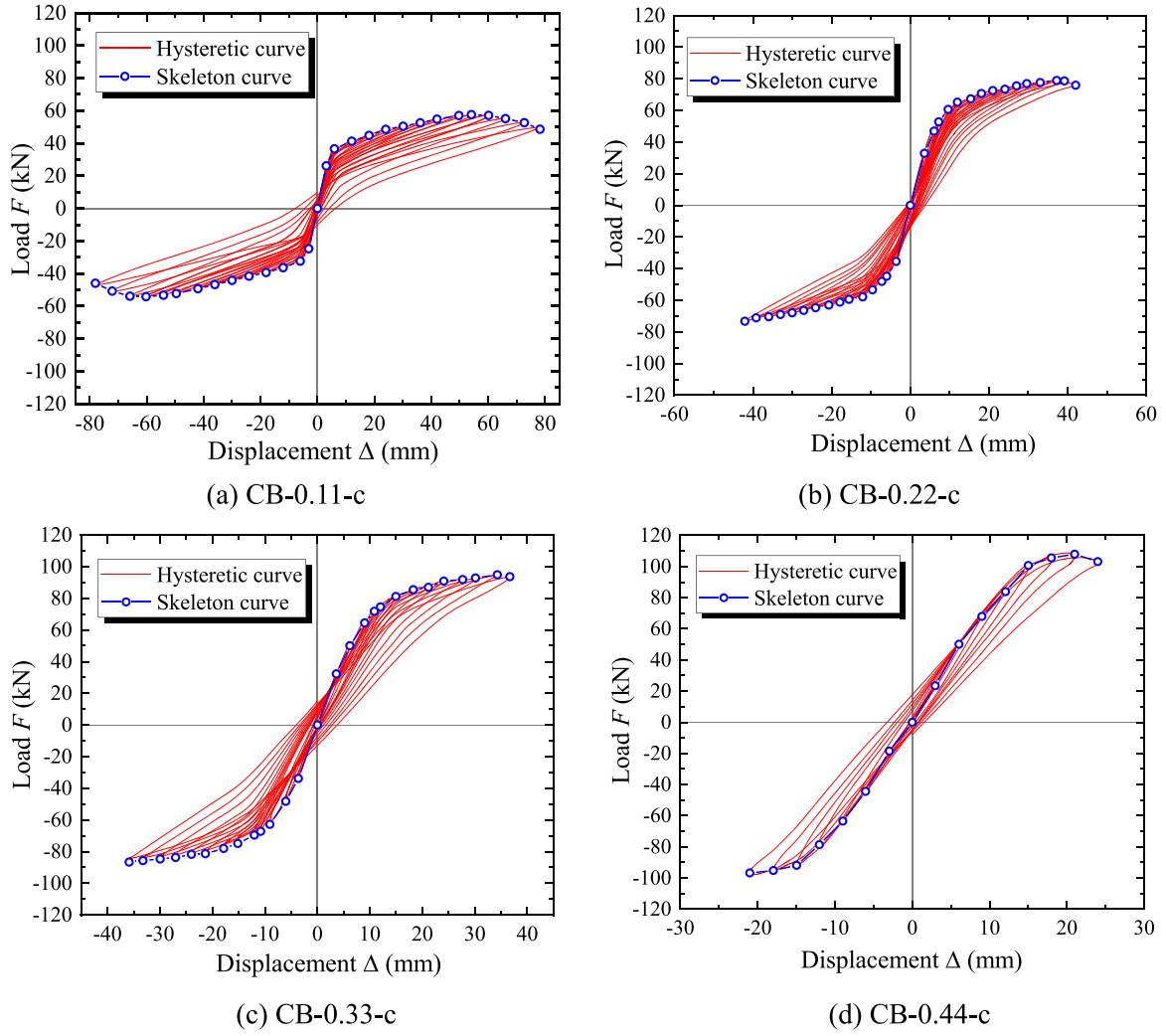


Fig. 9. Hysteretic curves of specimens.

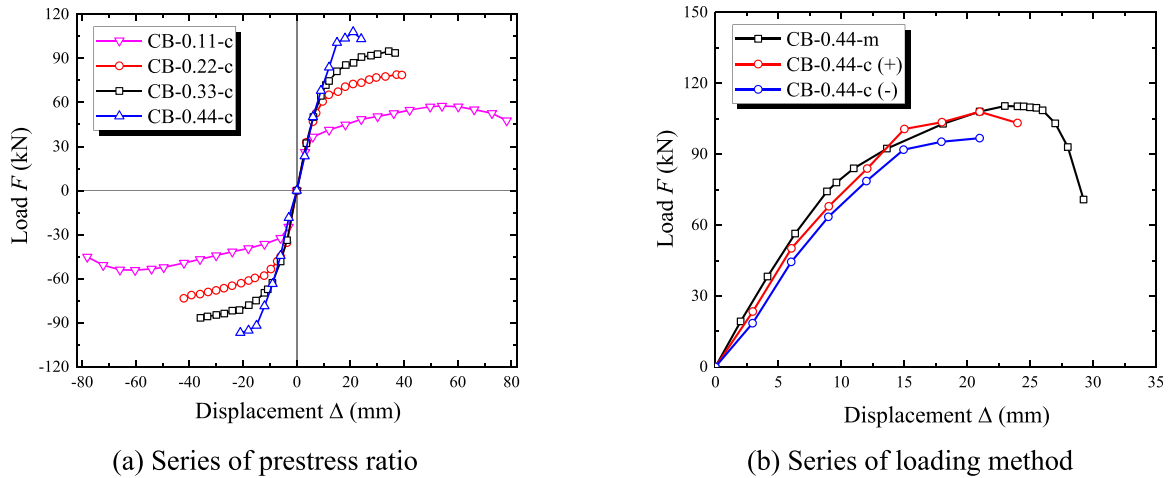


Fig. 10. Skeleton curves of specimens.

indicating that the stiffness before the joint opening was determined solely by the section's flexural stiffness, not controlled by the prestress. The difference in the stiffness changing points of specimens with different prestress ratios was significant. Specimens with higher prestress ratios could sustain more load growth along the initial

stiffness, while those with lower prestress ratios reached the second stiffness stage earlier. This indicates that the joint contact states of specimens with different prestress ratios were distinct, leading to significant differences in their bending capacity and ductility. In contrast, the curves of specimens with different loading methods showed little

Table 4
Characteristic points of specimens.

Specimen	Loading direction	Yield point		Peak point		Ultimate point		CI	DI
		F (kN)	Δ (mm)	F (kN)	Δ (mm)	F (kN)	Δ (mm)		
CB-0.11-c	Positive+	40.02	10.44	57.61	60.17	47.64	78.24	0.73	7.49
	Negative-	35.67	11.00	54.34	60.20	45.15	78.04	0.74	7.09
	Average	37.84	10.72	55.98	60.19	46.40	78.14	0.73	7.29
CB-0.22-c	Positive+	65.93	12.96	79.40	37.23	75.88	42.06	1.00	3.25
	Negative-	58.76	14.01	73.33	42.06	73.33	42.06	1.00	3.00
	Average	62.34	13.48	76.37	39.64	74.60	42.06	1.00	3.12
CB-0.33-c	Positive+	80.36	14.89	94.89	34.30	93.64	36.68	1.20	2.46
	Negative-	70.96	13.73	86.64	35.85	86.64	35.85	1.18	2.61
	Average	75.66	14.31	90.77	35.07	90.14	36.26	1.19	2.53
CB-0.44-c	Positive+	101.46	15.88	108.73	24.01	108.73	24.01	1.37	1.51
	Negative-	90.62	14.67	97.89	21.02	97.89	21.02	1.33	1.43
	Average	96.04	15.27	103.31	22.52	103.31	22.52	1.35	1.47
CB-0.44-m	/	95.22	14.85	110.35	23.02	80.94	26.94	1.97	1.73

variation.

3.3.1. Capacity index

To compare the differences in ductility and ultimate capacity among the specimens in detail, two indexes are introduced: the ultimate capacity index *CI* and the ductility index *DI*:

$$CI = \frac{F_u}{F_{ub}} \quad (1)$$

$$DI = \frac{\Delta_u}{\Delta_y} \quad (2)$$

Where, F_{ub} is the peak load of the benchmark specimen, which is taken as the specimen CB-0.22-c; Δ_y and Δ_u are the displacements of yield point and ultimate point, respectively.

The ultimate capacity indexes are listed in Table 4, and the average ultimate capacity indexes are used for comparison, as shown in Fig. 11. It can be found that the ultimate capacity of specimens was enhanced gradually with the increase of the prestress ratios. The specimen with the highest prestress ratio CB-0.44-c, showed an increase of 84.56 % compared to the specimen with the lowest prestress ratio CB-0.11-c. The prestress constraint the opening of the joint, allowing the bottom of the specimen to have a larger contact area to resist bending moments. This enabled a more effective utilization of the compressive strength of the concrete, resulting in a significant enhancement in ultimate capacity. Monotonic loading specimen exhibited higher ultimate capacity than cyclic loading specimen with the same prestress ratio. Specifically, the ultimate capacity of specimen CB-0.44-m was 6.81% greater than that of specimen CB-0.44-c. It can be explained that the cyclic loading specimen experienced cumulative damage in the process of compression and

decompression, resulting in a lower ultimate capacity. However, the ultimate capacity in positive direction of CB-0.44-c is only 1.49 % lower than that of CB-0.44-m, indicating that the overall difference is not significant.

3.3.2. Ductility index

The average ductility index of each specimen is shown in Fig. 12. It can be found that the ductility of specimens significantly increased as the prestress ratio decreased. The specimen with the lowest prestress CB-0.11-c, exhibited a 394.39 % increase in ductility compared to the specimen with the highest prestress CB-0.44-c. This is contrary to the trend of the ultimate capacity. This indicates that the smaller concrete compressive stress provides sufficient deformation capacity for the specimens with lower prestress ratio, allowing the tower to undergo larger cyclic rotations around the horizontal joint without damage. For specimens with different loading methods, the *DI* of specimen CB-0.44-m increased by 17.36 % compared to the *DI* of specimen CB-0.44-c, which is more pronounced than the comparison of ultimate capacity. Combined with the skeleton curves, it can be inferred that the cyclic loading specimens experienced cumulative damage, leading to rapid concrete crush after reaching the peak load. While the monotonic loading specimens exhibited a noticeable load drop stage, resulting in slightly improved ductility. But overall, both specimens exhibited poor ductility and experienced rapid failure after yielding due to the excessive prestress.

3.4. Stiffness degradation

The secant stiffness of the hysteresis loops gradually decreased with

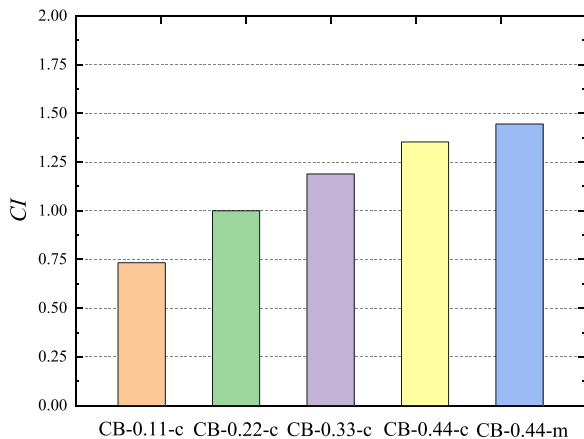


Fig. 11. Comparison of ultimate capacity indexes.

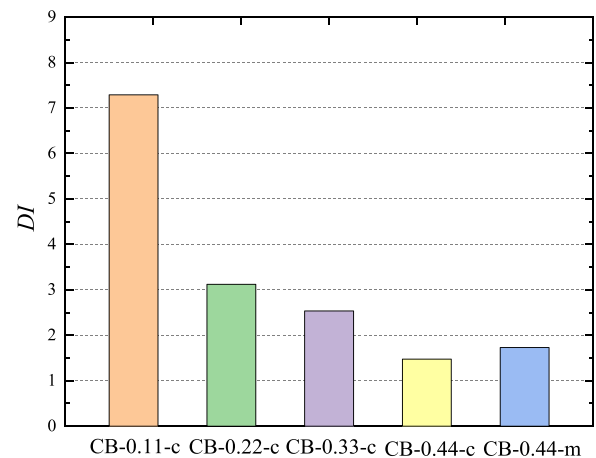


Fig. 12. Comparison of ductility indexes.

the increasing number of cycles, demonstrating a stiffness degradation phenomenon. The secant stiffness K_i was introduced to analyze the stiffness degradation patterns of different specimens, and the secant stiffness of specimens are showed in Fig. 13. The secant stiffness K_i is determined by:

$$K_i = \frac{|+F_i| + |-F_i|}{|+\Delta_i| + |-\Delta_i|} \quad (3)$$

Where, $+F_i$ and $-F_i$ are the peak loads in the i -th loading cycle of positive and negative direction, respectively; $+\Delta_i$ and $-\Delta_i$ are the displacements corresponding to $+F_i$ and $-F_i$.

It can be observed that specimens with higher prestress ratios have only slightly greater overall stiffness, further confirming that the initial stiffness of the specimens is not affected by the prestress. The specimen CB-0.11-c has significantly lower initial stiffness due to its low prestress. The horizontal joint opened during the loading cycle at ± 6 mm, leading to a rapid stiffness degradation and entering the second stiffness stage. However, the initial stiffness did not differ significantly from the other specimens.

In the early loading stage, the stiffness of specimens rapidly decreased. This was mainly caused by the opening of the horizontal joint, which increased the deformation capacity of the tower, resulting in a significant reduction in stiffness. However, after the horizontal joint closed, the stiffness could recover. As the load increased, the compression area of concrete decreased, leading to further reduction in stiffness. After the concrete on the compressive side reached the ultimate strength, the stiffness degradation gradually slowed down, indicating that the joint opening had fully developed. Specimens with lower prestress ratios exhibited a longer duration in this stage, corresponding to better ductility. For specimens with higher prestress ratios, this phenomenon was less pronounced, and the stiffness degradation rate remained relatively large. Additionally, all specimens with high prestress ratios exhibited a sharp degradation in stiffness after reaching the peak load, as indicated by the dashed lines in Fig. 13. This indicates that after reaching the peak load, the load did not show a gradual decline but instead dropped rapidly due to the sudden concrete crushing.

The specimen with the highest prestress ratio CB-0.44-c, exhibited a significantly different stiffness degradation pattern. In the early loading stage, the stiffness degradation was less than that of other specimens, indicating that the higher prestress restrained the joint opening, thereby reducing the specimen's deformation capacity. However, as loading progressed, the stiffness degradation rate increased sharply, which was completely opposite to the trend observed in other specimens. This can be attributed to that the joint opening was difficult to develop due to the excessively high prestress, resulting in high compressive stress in

concrete. Therefore, the concrete crushed suddenly, causing rapid stiffness degradation before reaching the peak load.

3.5. Energy dissipation capacity

The enclosed loop area in the hysteresis curve can represent the energy dissipation capacity of specimens during that loading cycle. The larger the enclosed area, the stronger the energy dissipation capacity of the specimen. According to calculation methods from [36–40], the energy dissipation E_d can be obtained, as shown in Fig. 14. Additionally, the equivalent viscous damping coefficient h_e [41] was adopted to further reflect the degree of plastic development of the specimens:

$$h_e = \frac{1}{2\pi} \frac{S_1}{S_2} \quad (4)$$

Where, S_1 is the enclosed area of the hysteresis loop, representing the total energy dissipated during that loading cycle; S_2 is the nominal elastic potential energy corresponding to the maximum displacement point of that loading cycle.

The variation of the equivalent viscous damping coefficients h_e with horizontal displacement Δ is shown in Fig. 15. In the early loading stage, the equivalent viscous damping coefficient decreased rapidly. This is attributed to that initial hysteresis loops had some areas due to the virtual displacements caused by installation errors, while the specimens still remained in the elastic stage with low nominal elastic potential energy. Consequently, the calculated equivalent viscous damping coefficients were larger. However, the energy dissipation of each specimen still increased with the horizontal displacement. As loading progressed, the influence of virtual displacements diminished, and the variation of h_e returned to normal.

As can be found from Figs. 14 and 15, the energy dissipation capacity of each specimen near the ultimate displacement showed little difference, but the energy dissipation of specimens with lower prestress is significantly less at the same displacement, indicating stronger ductility. Before reaching the peak load, the energy dissipation of specimen CB-0.11-c was relatively small, and the equivalent viscous damping coefficient remained nearly constant. This is because specimens with lower prestress ratios could sufficiently rotate around horizontal joints. Considering the concrete compressive stress is still small, this rotation resembled rigid body rotation, resulting in minimal plastic deformation. Therefore, the energy dissipation was low, and the specimens exhibited strong self-centering performance. After the specimen reached the peak load, the compressive concrete reached the ultimate compressive strain and crush, resulting in significant energy dissipation. It was found from the test phenomena that the concrete of specimen CB-0.11-c gradually spalled, leading to a slow decrease in load. Therefore, although the

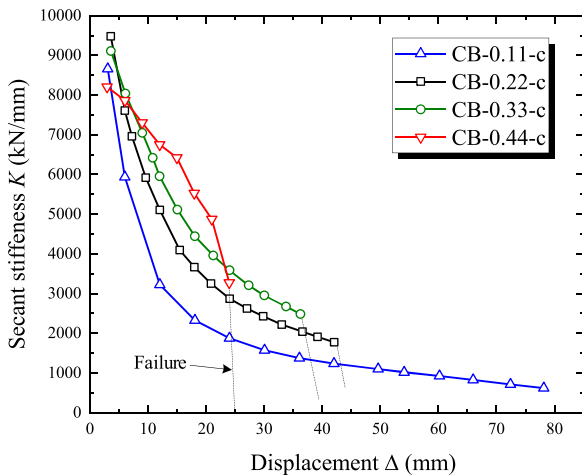


Fig. 13. Stiffness degradation curves of specimens.

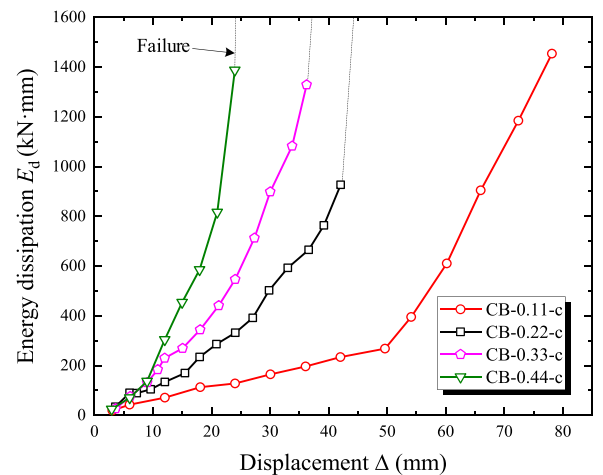


Fig. 14. Equivalent dissipation of specimens.

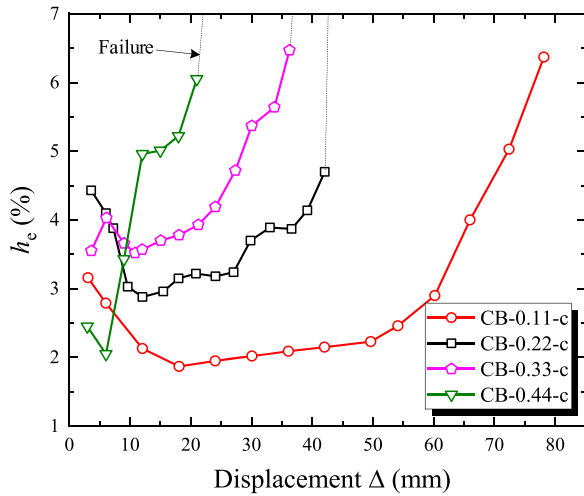


Fig. 15. Equivalent viscous damping coefficients of specimens.

equivalent viscous damping coefficient in the later stages was considerable, the gradual energy dissipation ensures the ductility of the specimen.

For specimens with higher prestress ratios, the initial energy dissipation and equivalent viscous damping coefficient increased rapidly in the early loading stages. This indicates that higher prestress could restrain joint opening, thus the specimen could be more like a cast-in structure, leading to gradual irreversible damage with the increase of the load. Combined with the test phenomenon, specimens with high prestress ratios experienced sudden concrete crushing within the first two loading cycles after reaching peak load, resulting in a rapid drop in load and brittle failure of the specimen. Thus, the gradual energy dissipation could hardly be found on specimens with high prestress ratios before reaching peak load, but significant energy dissipation suddenly occurred at the ultimate displacement, indicating excessive and irreparable damage to the structure. The greater the prestress ratio, the more pronounced the above trend, resulting in weaker self-centering capability of the specimen.

3.6. Residual displacement

The residual displacement Δ_r is the irreversible displacement of the specimen when unloaded to zero at each loading cycle. Compared to energy dissipation capacity, residual displacement can more directly describe the specimen's self-centering ability. The average residual displacement of positive and negative directions of each specimen are shown in Fig. 16.

All specimens showed a gradual increase in residual displacement with loading, but those with lower prestress ratios had significantly smaller residual displacements. The residual displacement of specimen CB-0.11-c was at a relatively low level and remained almost unchanged during the early loading stage. At the peak point, the residual displacement only increased by 1.72 mm compared to that at the first loading cycle, while the horizontal displacement had already increased by 60.17 mm. The specimen demonstrated good self-centering performance, allowing it to continue bearing load while the horizontal joint repeatedly opened and closed. After reaching the peak load, as the concrete began to crush, the specimen started accumulating plastic deformation, leading to a rapid increase in residual displacement and a reduction in self-centering performance. For specimens with higher prestress, there was a noticeable increase in residual displacement during the early loading stage, with a more rapid increase as they approached the peak load. Specimen CB-0.44-c exhibited a rapid and linear increase in residual displacement from the early loading, indicating that the specimen continuously accumulated irreversible

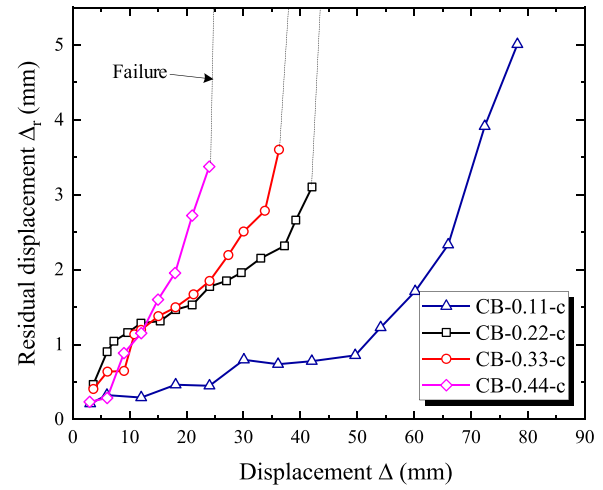


Fig. 16. Residual displacement of specimens.

deformation, which aligns with the conclusions regarding energy dissipation capacity.

It can be observed that the residual displacement of the specimen CB-0.11-c was large due to the large overall displacement. But this does not indicate that the irreversible deformation of specimen CB-0.11-c was greater. Therefore, the residual displacement ratio λ_r [42,43] is introduced to compare residual displacement levels. As a dimensionless index, λ_r can more directly describe the self-centering performance of specimens:

$$\lambda_r = \frac{\Delta_{p,r}}{\Delta_p} \quad (5)$$

The comparison of the residual displacement ratios of specimens are shown in Fig. 17. It can be observed that the residual displacement ratio significantly decreased with the reduction of the prestress ratio. This indicates that specimens with lower prestress ratios exhibited minimal irreversible deformation before reaching the peak load, demonstrating a strong self-centering capability, which was beneficial for damage retrofitting. Specimens with higher prestress ratios continuously accumulated plastic damage during loading before failure. Although the absolute value of residual displacement was relatively small, the residual displacement ratio was significant, indicating weaker self-centering performance.

Combining the conclusions of energy dissipation capacity and residual displacement, it indicates that the tower with lower prestress ratio

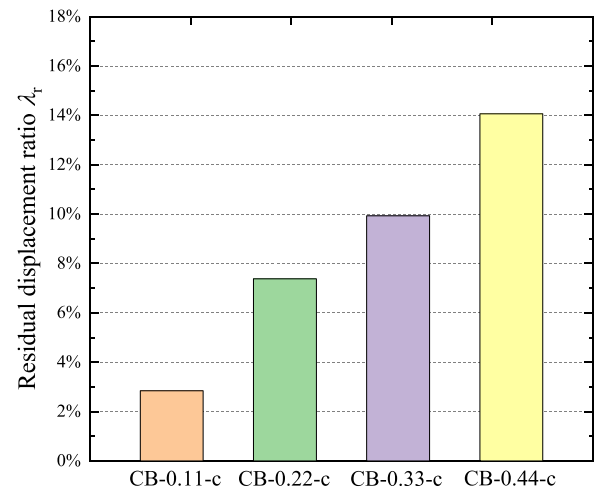


Fig. 17. Residual displacement ratios of specimens.

can more freely rotate around the joint, resulting in smaller irreversible deformations, stronger self-centering performance, and greater reparability after extreme load damage.

3.7. Prestress variation

The variations of prestress values σ_p for the eight tendons of specimens are shown in Fig. 18. Overall, the prestress increased continuously with the progression of cyclic loading. This is attributed to that, as the horizontal displacement increased, the tower rotated around the joint after the joint opened, and the tendons on the tensile zone were elongated due to the rotation, resulting the increase in prestress. In the

meanwhile, the compressive deformation occurred on the compression zone of the tower due to the bending moment, leading to the shortening of the tendons, which resulted in a reduction of the prestress of the tendons on the compressive side. Therefore, during positive loading, the prestress in the tendons P1, P2, and P8 located on the compressive side decreased, while the prestress in the tendons P4, P5, and P6 located on the tensile side increased. The pattern is reversed during negative loading. Regardless of positive or negative loading, the prestress in the tendons P3 and P7, which are located on the centroidal axis of the section, consistently increased with the increase in horizontal displacement. However, as the prestress ratio increased, the increments of prestress in P3 and P7 significantly decreased. This indicates that the

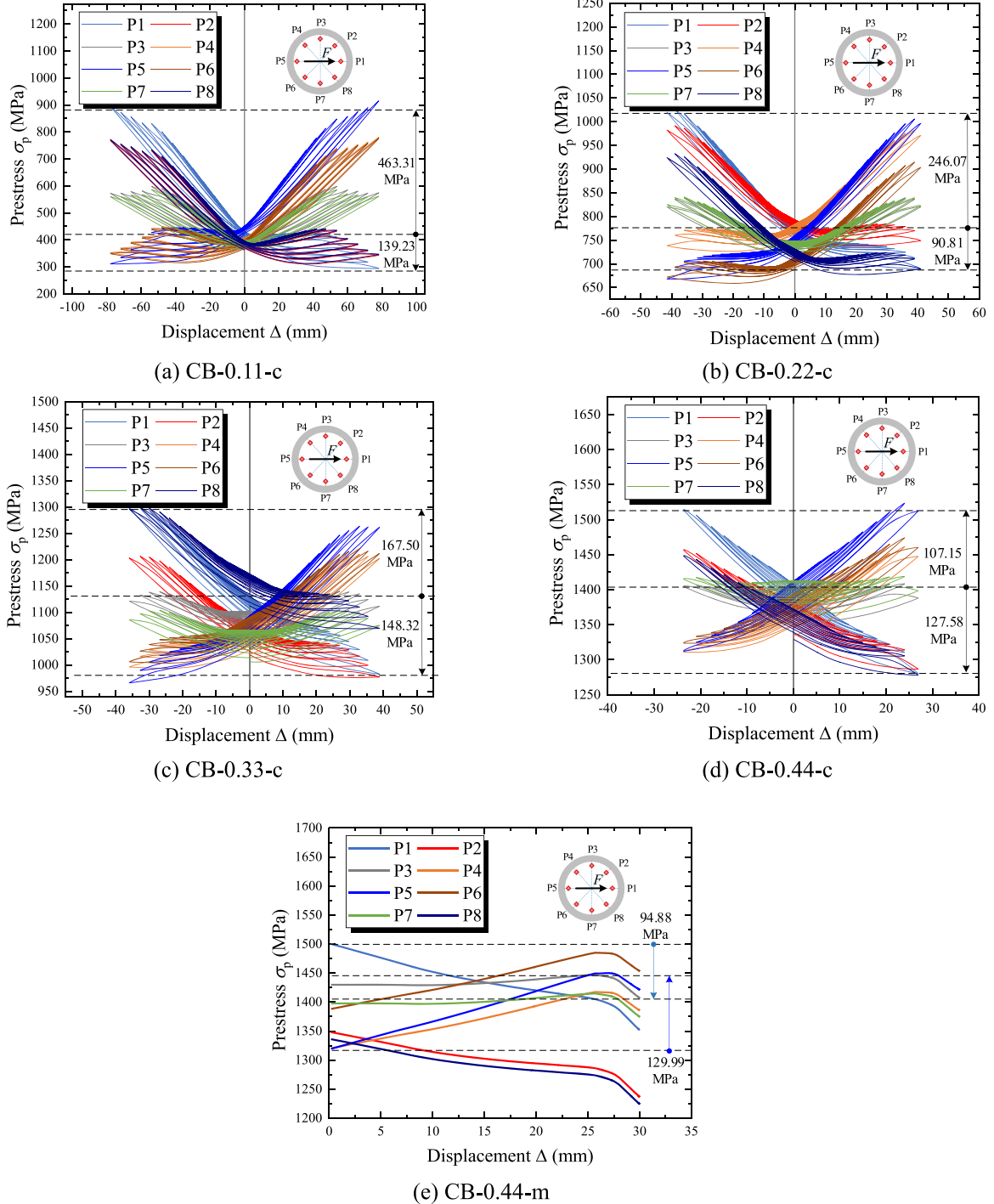


Fig. 18. Prestress of specimens.

increase in prestress ratio made the neutral axis be closer to the centroidal axis, resulting in a larger sectional area being in a compressive state. The prestress variations of P1 and P5 were the largest during negative and positive loading, respectively, due to the largest distances of P1 and P5 from the neutral axis. In contrast, the prestress increments of the tendons closer to the neutral axis were much smaller.

Taking the tendon P1 as an example, the comparison of prestress variation for different specimens is indicated in Fig. 18. The prestress increment of P1 in specimen CB-0.44-c decreased by 76.87% compared to specimen CB-0.11-c, indicating that the prestress increment significantly decreases as the prestress ratio increases. This is attributed to that the boundary condition at the bottom of the specimen became more stable as the prestress ratio increased, weakening the ability of joint opening. Consequently, the elongation of the tendons decreased significantly, leading to a reduction in the prestress increment. On the contrary, the smaller prestress ratio enhanced the deformation capacity of the specimen, significantly increasing the prestress increment. As the prestress ratio increased, the increments of prestress of P1 during negative loading and the decrements of prestress of P1 during positive loading became closer. This further indicated that the increase in prestress ratio moved the neutral axis closer to the centroidal axis, resulting in similar length variations of the tendons during loading in both directions. The prestress variation pattern could be observed more directly through specimen CB-0.44-m, as shown in Fig. 18(e). The prestress varied linearly before reaching the peak load, indicating a linear relationship between the length variation of the tendons and the horizontal displacement. However, a sudden large compression deformation occurred after reaching the peak load due to the concrete crush at the bottom of the specimen, leading to the shortening of all the tendons, resulting in a sharp decrease in prestress. The prestress further decreased as the concrete continued to crush, corresponding to the degradation process of the load-bearing capacity.

In addition to the rapid decrease in prestress due to significant compressive deformation at failure, the prestress in cyclic loading specimens were also reduced after multiple cycles of loading before failure. From the prestress hysteretic curves of tendons in Fig. 18, it can be observed that the prestress corresponding to the horizontal displacement at zero decreased gradually with an increase in the number of cyclic loops, indicating a loss of prestress. The average prestress loss and prestress loss ratio for different specimens are shown in Fig. 19. It can be observed that, at the same horizontal displacement, specimens with a smaller prestress ratio experienced significantly less prestress loss. Compared to specimen CB-0.44-c, specimen CB-0.11-c showed a 97.52 % reduction in prestress loss and a 91.80 % reduction in the corresponding prestress loss ratio. According to the conclusion in Section 3.5, specimens with larger prestress ratios exhibited irreversible

deformations in the early loading stage. Therefore, the pronounced significant energy dissipation can be found in Fig. 19, leading to a significant prestress loss in the early loading process. As the horizontal displacement increased, more severe compressive damage occurred at the bottom of the specimens, resulting in a more significant increase in prestress loss. For specimens with lower prestress ratios, the prestress loss during the early loading phase was slight. As discussed in Sections 3.5 and 3.6, this is attributed to their stronger self-centering performance, thus the irreversible compressive deformation in the early stages was minimal. However, specimens with lower prestress ratios experienced gradual energy dissipation during the later loading phase, leading to a gradually increasing prestress loss. Thus, the prestress loss of specimen CB-0.11-c during the later loading stage became comparable to that of specimen CB-0.44-c. Furthermore, due to its initial lower prestress ratio, the prestress loss ratio increased more significantly, as shown in Fig. 19(b). This further reflects that the tower exhibits strong self-centering performances and shows ductile failure characteristics when the prestress ratio is low.

4. Evaluation of existing design methods

Practical calculation method for the ultimate bending capacity of prestressed concrete towers is still rare. But corresponding engineering design methods with reference to material mechanics and design codes of building structures can be found, which can be used for preliminary predictions of the ultimate bending capacity of towers. The accuracy and applicability of two commonly used engineering design methods were evaluated based on the experimental results.

4.1. Method based on plane-section assumption

The plane-section assumption can simplify the stress calculation when combined with the linear elastic assumption. It is commonly applied in the strength verification of beams and columns. After the joint opened in prestressed concrete towers, the plane-section assumption is not adequately satisfied at the bottom of the towers. Therefore, the existing design method conservatively limits the joint from opening under bending moments, meaning the tensile stress of the joint does not exceed the tensile strength. Thus, the concrete compressive strength can be verified based on the plane-section assumption.

$$M = \min \left\{ \left(f_{tc} + \frac{P+N}{A_c} \right) \cdot W_c, \left(f_c - \frac{P+N}{A_c} \right) \cdot W_c \right\} \quad (6)$$

Where, f_{tc} is the tensile strength of concrete; P and N are total prestressing force and vertical load, respectively; A_c and W_c are section area and section modulus.

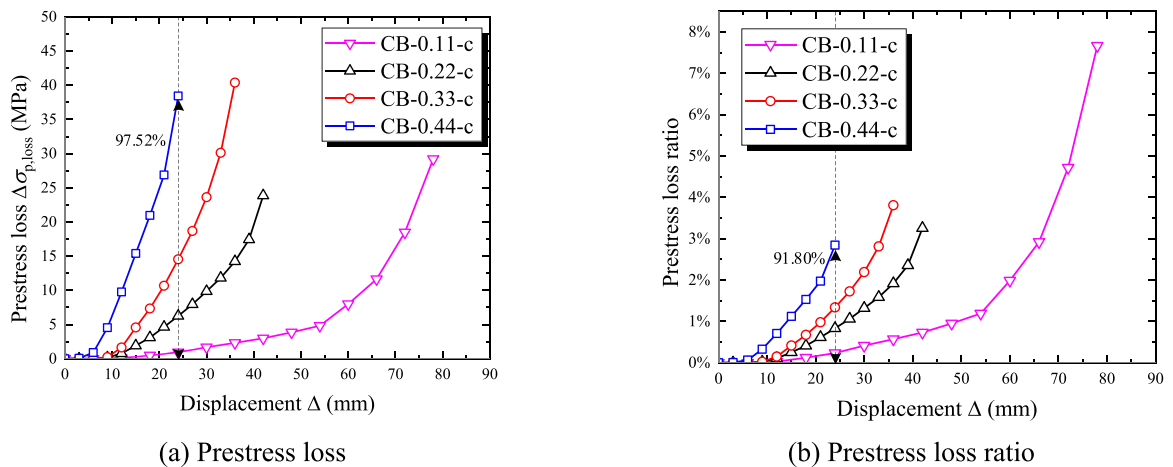


Fig. 19. Prestress loss of specimens.

Based on Eq. (6), the ultimate bending capacity of each specimen was calculated and converted to the lateral force F_{uc1} , with the results shown in Table 5. It can be observed that this method significantly underestimated the ultimate bending capacity of all specimens. This method adopts elastic strength verification by assuming that the joint does not open. However, it was evident that the bending moment could continue to increase even after the joint opened, and the bending moment at the state when joint opened is smaller compared to the ultimate moment. The difference between the calculated moment and the ultimate moment is greater when the prestressing force is lower. Therefore, this design method is overly conservative, and the calculated moment can only be considered as the yielding moment rather than the ultimate moment. Moreover, the cross-section deformation no longer conforms to the plane-section assumption after the joint opened, making this method inapplicable. But for towers with high safety redundancy requirements, this method can still be adopted to conservatively check the bending capacity.

4.2. Method based on GB50135-2019

GB50135-2019 [44] specifies the compression-bending capacity for cast-in-place prestressed concrete towers. This method considers the contributions of prestressed tendons and reinforcement bars, simplifying the trapezoidal stress distribution of the cross-section to an equivalent rectangular stress distribution. In this method, the state when both prestressed tendons and reinforcement bars yield are defined as the ultimate state of towers, and the calculation formulas based on this ultimate equilibrium condition are provided:

$$N \leq \alpha \alpha_1 f_c A_c - \sigma_{p0} A_p + \alpha f_{py} A_p - \alpha_t (f_{py} - \sigma_{p0}) A_p + (\alpha - \alpha_t) f_y A_s \quad (7)$$

$$M \leq \left(\alpha_1 f_c A_c r_0 + r_0 f_y A_s + r_p f_{py} A_p \right) \frac{\sin \alpha \pi}{\pi} + \left[(f_{py} - \sigma_{p0}) A_p r_p + r_0 f_y A_s \right] \frac{\sin \alpha_t \pi}{\pi} \quad (8)$$

Where, f_y and A_s are the strength and section area of vertical reinforcement bars. Since the tower reinforcement at the joint is not continuous, the strength of reinforcement bars cannot contribute to the ultimate capacity when the joint opened under large bending moment. Therefore, the reinforcement term in the above equation should not be considered. In addition, f_{py} is the compressive strength of prestress tendons. According to GB50011-2010 [45], the compressive strength of external unbonded prestressing tendons should not be considered. Thus, the above equations should be simplified to:

$$N \leq \alpha \alpha_1 f_c A_c - \sigma_{p0} A_p - \alpha_t (f_{py} - \sigma_{p0}) A_p \quad (9)$$

$$M \leq \alpha_1 f_c A_c r_0 \frac{\sin \alpha \pi}{\pi} + (f_{py} - \sigma_{p0}) A_p r_p \frac{\sin \alpha_t \pi}{\pi} \quad (10)$$

Where, σ_{p0} is the initial prestress; f_{py} is the yield strength of the prestressing tendons; r_0 is the middle radius of the cross-section; r_p is the radius where the prestressing tendons are distributed; α is the half-angle coefficient of the compression zone, representing the area of the compression zone; α_t is the half-angle coefficient of the tensile reinforcement, calculated as $\alpha_t = 1 - 1.5\alpha$, with a minimum value of 0.

Based on Eqs. (9) and (10), the ultimate bending capacity of each specimen was calculated and converted to the lateral force F_{uc2} , with the results shown in Table 5. It can be seen that the design method based on GB50135-2019 significantly overestimated the ultimate bending capacity of towers with low prestress ratios, but underestimated that of towers with large prestress ratios. The reason is that the method is designed for cast-in-place towers, assuming that the ultimate state is characterized by both reinforcement bars and prestressing tendons yielding in tension. However, the joints may open for wind turbine prestressed concrete towers under bending, preventing the reinforcement bars from utilizing the strength. Besides, external unbonded tendons are typically designed to be not yielding under bending. The increase in stress up to yielding strength is greater when the initial prestress ratio is lower, resulting in a larger calculated bending moment. Therefore, the calculated results of towers with lower prestress ratios were higher, and the bending capacity of towers with higher prestress ratios were underestimated. It can be seen that this method is not suitable and will generally overestimate the ultimate bending capacity of the towers within the common range of prestress ratios.

5. Development of theoretical model

Due to the lack of effective calculation methods and theoretical analysis methods, FE solid models are generally used to calculate the ultimate capacity of towers. The calculation efficiency of FE analysis is low although the accuracy is high. Considering this, a theoretical analysis model was established to efficiently analyze the compression-bending behavior and ultimate capacity of towers.

5.1. Calculation assumptions

In cast-in-place concrete structures, the equivalent plastic hinge model is an important method for analyzing the elastic-plastic behavior [46]. It assumes that a plastic hinge develops within a certain length range after yielding occurred in the structure, and subsequent plastic deformation and damage will concentrate in that plastic hinge region. According to this model, the curvature of the tower is composed of elastic and plastic curvature. The elastic curvature is linearly distributed along the height of the tower, while the plastic curvature is concentrated in the region of the most severe bending, specifically at the bottom of the tower. The actual plastic hinge rotation can be equated to the parallelogram area at the bottom of the tower, as shown in Fig. 20. The equivalent plastic hinge rotation can be calculated by the following equation:

Table 5
Comparison between the calculated and measured ultimate capacity of specimens.

Specimen label	Loading direction	F_{ue} (kN)	F_{uc1} (kN)	F_{uc1} / F_{ue}	F_{uc2} (kN)	F_{uc2} / F_{ue}	F_{uc3} (kN)	F_{uc3} / F_{ue}
CB-0.11-c	Positive+	57.61	24.56	0.426	115.52	2.005	55.62	0.965
	Negative-	54.34	24.56	0.452	115.52	2.126	55.62	1.024
	Average	55.98	24.56	0.439	115.52	2.064	55.62	0.994
CB-0.22-c	Positive+	79.40	36.84	0.464	111.26	1.401	76.45	0.963
	Negative-	73.33	36.84	0.502	111.26	1.517	76.45	1.043
	Average	76.37	36.84	0.482	111.26	1.457	76.45	1.001
CB-0.33-c	Positive+	94.89	49.01	0.517	95.68	1.008	89.94	0.948
	Negative-	86.64	49.01	0.566	95.68	1.104	89.94	1.038
	Average	90.77	49.01	0.540	95.68	1.054	89.94	0.991
CB-0.44-c	Positive+	108.73	59.24	0.545	73.81	0.679	96.38	0.886
	Negative-	97.89	59.24	0.605	73.81	0.754	96.38	0.985
	Average	103.31	59.24	0.573	73.81	0.714	96.38	0.933
CB-0.44-m	/	110.35	58.90	0.534	74.69	0.677	96.12	0.871

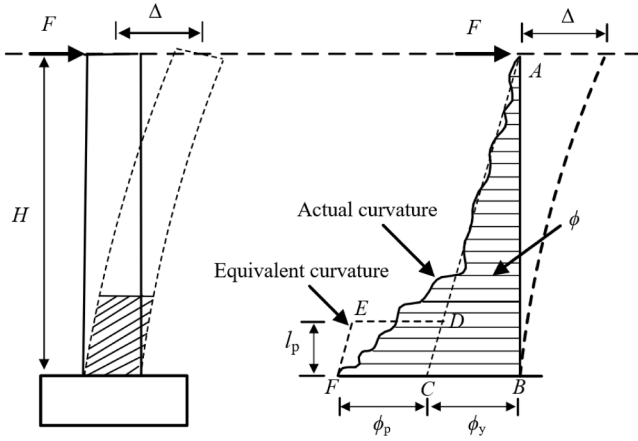


Fig. 20. Equivalent plastic hinge model.

$$\theta_{tp} = (\phi - \phi_y)l_p \quad (11)$$

The equivalent plastic hinge in cast-in-place structures is generated due to the structural plastic deformation. On the other hand, the deformation of prestressed concrete towers mainly consists of the rigid body rotation and the elastic deformation of the tower segment. Although the plastic deformation is minimal, the rotation generated by the joint opening leads to significant nonlinear deformation. Therefore, the nonlinear behavior of joint opening can be regarded as an equivalent plastic hinge concentrated at the bottom of the tower, allowing for analysis adopting the equivalent plastic hinge model. The following basic assumptions were proposed based on test results to simplify the theoretical model:

- (1) The failure of the tower only occurs at the bottom section. According to the test results, the bottom section near the joint was the critical section, thus only the bottom section was used for moment-curvature analysis.

- (2) The lateral displacement is composed of the elastic deflection of the tower and the displacement caused by rotation around the joint. Due to the lack of reinforcement at the joint, the tower stress is minimal except for the bottom area near joint. Therefore, the plastic deformation of the tower was neglected.
- (3) The load-bearing contribution of the reinforcement is neglected, and the concrete tensile strength is ignored. Since the reinforcement bars are not continuous, the tensile strength of reinforcement and concrete cannot be utilized.
- (4) Only the variation in prestress caused by the joint opening is considered, ignoring the prestress losses caused by the tower deflection and other effects. According to the test results, the prestress varied linearly before reaching the peak load, thus it is assumed that the prestress variations remain within the elastic range.
- (5) The compressive strain of the concrete in the compression zone is approximately distributed linearly. Besides, the length variations of prestressed tendons were approximately linear before the tower reached the ultimate state.

5.2. Geometric deformation conditions

The geometric conditions of the prestressed concrete towers under bending can be defined based on the deformation characteristics at different stages. Based on the test results, the stress characteristics of the tower was significantly changed after the joint opened. Therefore, the overall geometric deformation conditions can be described using a two-stage characteristic model, as shown in Fig. 21.

5.2.1. Stage before the joint opened

In this stage, the joint has not yet opened, and the bottom section was in tight contact with the foundation. As the lateral force increased, the compressive strain on the tensile side of the bottom section gradually decreased until the compressive strain reached zero, at which point the joint opened. According to the curvature-strain relationship, the sectional curvature ϕ_1 at this point can be calculated as:

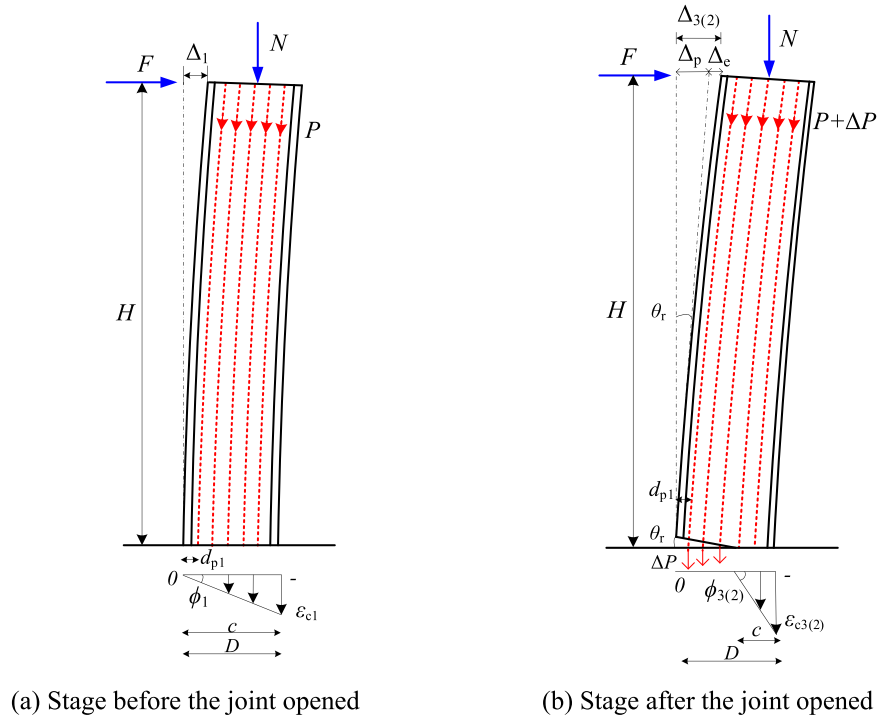


Fig. 21. Schematic diagram of geometric deformation of prestressed concrete tower.

$$\phi_1 = \frac{\varepsilon_{c1}}{D} \quad (12)$$

Where, ε_{c1} is the compressive strain on the compressive side of the bottom section at this state.

Since the joint has not yet opened, the stress pattern of the tower is similar to that of a cast-in-place cantilever column. Thus, the top horizontal displacement was caused only by the deflection of the tower, as shown in Fig. 21. According to the Euler-Bernoulli beam theory, the top horizontal displacement Δ_1 of the tower when the joint opens can be calculated as:

$$\Delta_1 = \frac{1}{3}\phi_1 H^2 \quad (13)$$

5.2.2. Stage after the joint opened

After the joint opened, the tower began to rotate around the joint while undergoing tower deflection. Due to the variations of the height of the compression zone and the elongation of the prestress tendons, the deformation in this stage exhibited significant nonlinear characteristics even if the concrete remained in the elastic stage. Therefore, the joint rotation θ_r is equivalent to the concentrated plastic hinge θ_{tp} at the bottom of the tower by adopting the equivalent plastic hinge model.

$$\theta_{\text{tp}} = \theta_r \quad (14)$$

At this stage, the horizontal displacement Δ was caused by both the tower deflection and the rotation around the joint, as shown in Fig. 21. The deflection resulted in elastic displacement Δ_e , while the rotation led to plastic displacement Δ_p :

$$\Delta = \Delta_e + \Delta_p \quad (15)$$

The height of the compression zone c continuously decreased during the process of joint opening, leading to the elongation of the tensile-side tendons and the shortening of the compressive-side tendons. As can be found from Fig. 21, all prestressed tendons remained in the compression zone when the joint opened to the position of the first prestressed tendon d_{p1} . Therefore, no increase of prestress was considered before this state when the effect of tower deflection is ignored. It can be assumed that the prestress only began to vary after the joint opened to the position of the first prestressed tendon. After the prestress began to change, the stiffness of the tower would be different. Thus, the displacement at this state should be calculated separately. The stage after the joint opened could be further subdivided into two stages (stage 2 and stage 3). The geometric deformation conditions of these two stages were similar, but the variation of prestress must be considered in stage 3, leading to differences in stiffness between stage 2 and stage 3. According to Eq. (15), the top horizontal displacement Δ_2 consisted of Δ_{2e} and Δ_{2p} . Δ_{2e} is the elastic displacement caused by the tower deflection, and the calculation method is the same as in stage 1. Therefore, it can be obtained by linearly extrapolating the displacement Δ_1 from stage 1:

$$\Delta_{2e} = \frac{F_2}{F_1} \Delta_1 \quad (16)$$

Where, F_1 is the lateral force of stage 1; F_2 is the lateral force of stage 2.

Δ_{2p} is the displacement caused by the joint rotation θ_r when it opened to that position. Based on the geometric relationship in Fig. 21 (b), it can be expressed as:

$$\Delta_{2p} = \theta_r H \quad (17)$$

Combined with Eqs. (11) and (14), it can be obtained that:

$$\Delta_{2p} = (\phi_2 - \phi_{v2})l_p H \quad (18)$$

Where, ϕ_{y2} is the yield curvature of stage 2, which refers to the portion of the total curvature composition excluding the plastic curvature. Similar to the elastic displacement Δ_{2e} , the yield curvature ϕ_{y2} is also caused by the tower deflection and can be linearly extrapolated from the

curvature ϕ_1 of stage 1. Thus, Eq. (18) can be further expressed as:

$$\Delta_{2p} = \left(\phi_2 - \frac{F_2}{F_1} \phi_1 \right) l_p H \quad (19)$$

l_p is the length of the equivalent plastic hinge. According to the research by Hewes [25], the length of the equivalent plastic hinge for external unbonded prestressed column depends only on the section dimension. Thus, it is taken as $0.5D$ based on the result by Hewes. The total curvature ϕ_2 in stage 2 can be determined by the compressive strain ε_{c2} at the end of the compression side:

$$\phi_2 = \frac{\varepsilon_{c2}}{d_{p,1}} \quad (20)$$

It can be found that the displacement of stage 2 is controlled by the distance $d_{p,1}$. Therefore, this stage can be neglected when $d_{p,1}$ is close to the diameter D , and the analysis can proceed directly to stage 3.

After the joint opened to the position of the first tendon, the tendons in the opened area began to elongate, while the tendons in the compression zone were shortened, as shown in Fig. 22. In Fig. 22, Δl_{pt+} represents the elongation of the tendons, and Δl_{pt-} represents the shortening of the tendons. According to the geometric relationship, the variation in tendon length $\Delta l_{pt,i}$ can be calculated as:

$$\Delta l_{\text{pt},i} = \theta_r(d_{\text{p},i} - c) \quad (21)$$

Where, $d_{p,i}$ is the distance from the tensile side of the tendon i to the edge of compressive side; c is height of the compression zone.

The horizontal displacement Δ_3 of the tower in stage 3 also consists of Δ_{3e} and Δ_{3p} . The elastic displacement Δ_{3e} can still be obtained by linearly extrapolating the displacement Δ_2 of stage 2:

$$\Delta_{3e} = \frac{F_3}{F_2} \Delta_2 \quad (22)$$

According to equivalent plastic hinge model, the equivalent plastic displacement Δ_{3p} can be obtained:

$$\Delta_{3p} = (\phi_3 - \phi_{v3}) l_p H \quad (23)$$

The yield curvature ϕ_{y3} of stage 3, including the elastic curvature caused by the tower deflection and the curvature of stage 2. Therefore, ϕ_{y3} can be obtained from linear extrapolation of the curvature ϕ_2 , making Eq. (23) transform into:

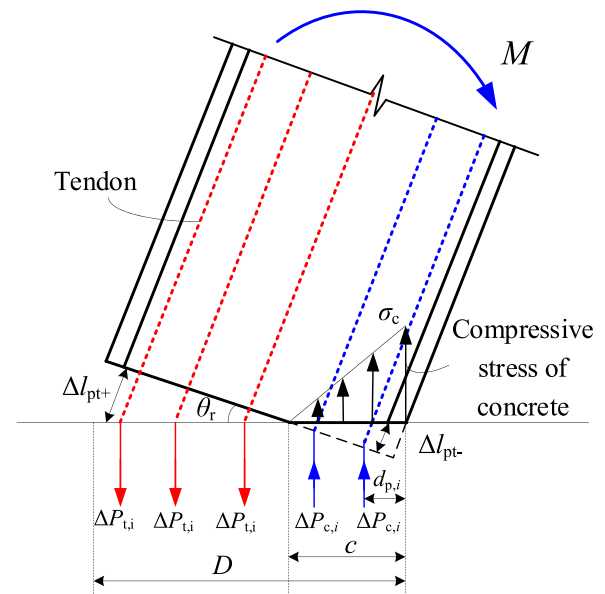


Fig. 22. Schematic diagram of force and deformation at the bottom of the tower in stage 3.

$$\Delta_{3p} = \left(\phi_3 - \frac{F_3}{F_2} \phi_2 \right) l_p H \quad (24)$$

The total curvature ϕ_3 in stage 3 can be determined by the compressive strain ε_{c3} at the edge of the compression zone on the concrete:

$$\phi_3 = \frac{\varepsilon_{c3}}{c} \quad (25)$$

The above equation is related to the height of the compression zone c , which varies with the increase of the rotation of the joint. Thus, each height of the compression zone corresponds to one curvature ϕ_3 and compressive strain ε_{c3} that satisfy the geometric equations.

5.3. Constitutive conditions

Based on the concrete constitutive model adopted in the derivation of the bending capacity of normal section in GB50010-2010 [45], the classical Rüschi curve [47] was introduced to describe the stress-strain relationship, as shown in Fig. 23 and following equations:

$$\sigma_c = \begin{cases} f'_c \left[2 \frac{\varepsilon_c}{\varepsilon_{c0}} - \left(\frac{\varepsilon_c}{\varepsilon_{c0}} \right)^2 \right] & \varepsilon_c \leq \varepsilon_{c0} \\ f'_c \varepsilon_{c0} & \varepsilon_c > \varepsilon_{c0} \end{cases} \quad (26)$$

It can be found that no decrease stage exists in the Rüschi curve after reaching the compressive strength. Considering the need for multiple iterations in analytical calculations, the calculation convergence of relationship without decrease stage would be better [48]. Besides, the descending behavior was not the focus since the ultimate bending capacity of the tower was primarily focused. Therefore, the ultimate state was defined as the state when the edge of compressive side reached the ultimate compressive strain ε_{cu} . This is also the reason why GB50010-2010 adopted the Rüschi curve as well. In addition, the effect of reinforcements was not considered in the Rüschi curve, which was more conservative for such thin-walled concrete structure.

Based on test results, the prestressed steel strands remained elastic state throughout the loading process. For simplification, the ideal elastoplastic model can be used to describe the stress-strain relationship of the prestress tendons, as shown in Fig. 23 and the following equation:

$$\sigma_{pt} = \begin{cases} E_p \varepsilon_{pt} & \varepsilon_{pt} \leq \varepsilon_{py} \\ f_{py} & \varepsilon_{pt} > \varepsilon_{py} \end{cases} \quad (27)$$

Since the concrete and tendons do not satisfy the deformation coordination condition, the prestress variation $\Delta\sigma_{pt,i}$ can only be calculated through the deformation relationship shown in Fig. 22. Combined with the stress-strain relationship and the deformation of the prestress tendons, the prestress variation $\Delta\sigma_{pt,i}$ can be obtained through the length variation Δl_{pt} :

$$\Delta\sigma_{pt,i} = \frac{(\phi_3 - \phi_2)(d_{p,i} - c)l_p}{l_{pt}} E_p \quad (28)$$

5.4. Equilibrium conditions

The stress and strain analysis of the bottom section after the joint opened was illustrated in Fig. 24. The compressive force of concrete F_c , initial prestressing force P , the variation of prestressing force ΔP , and the vertical load N remained in vertical equilibrium:

$$F_c = P + N + \sum_{i=1} \Delta P_i \quad (29)$$

As shown in Fig. 24, the compressive resultant force of concrete F_c was obtained by integrating the compressive stress of the concrete over the compression zone of the bottom section. According to Eq. (25) and the geometric relationship shown in Fig. 22, the compressive strain $\varepsilon_c(x)$ at the distance x from the section center was given by:

$$\varepsilon_c(x) = (x - R + c) \cdot \phi \quad (30)$$

Where, R is the outer radius of the cross-section.

The compressive force of concrete F_c was calculated by:

$$\begin{aligned} F_c &= \int_A \sigma_c dA = \int_A \sigma_c ((x - R + c) \cdot \phi) dA \\ &= \int_{-\theta_1}^{\theta_1} \int_r^R \sigma_c ((\rho \cos \theta - R + c) \cdot \phi) \cdot \rho d\rho d\theta \end{aligned} \quad (31)$$

$$\theta_1 = \arccos\left(\frac{R - c}{r_0}\right)$$

Where, θ_1 is the half arc angle of the compression zone; R , is the outer radius of the cross-section; r is the inner radius of the cross-section; r_0 is the middle radius of the cross-section, calculated as $r_0 = R - t/2$.

Tendons at different positions exhibited different variation of prestressing force ΔP , which were classified into prestress increments or prestress reductions based on the elongation or shortening of tendons. Combined with Eq. (27), the variation of prestressing force can be given by:

$$\Delta P_i = \frac{(\phi_3 - \phi_2)(d_{p,i} - c)l_p}{l_{pt}} E_p \cdot A_p \quad (32)$$

According to Eq. (25), the curvature ϕ corresponding to each height of the compression zone c can be obtained by combining the geometric deformation, the constitutive, and the equilibrium equations. The bending moment M can be obtained:

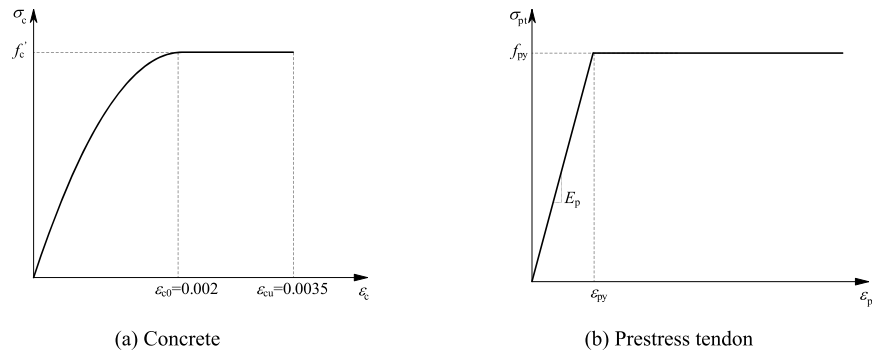


Fig. 23. Stress-strain relationships.

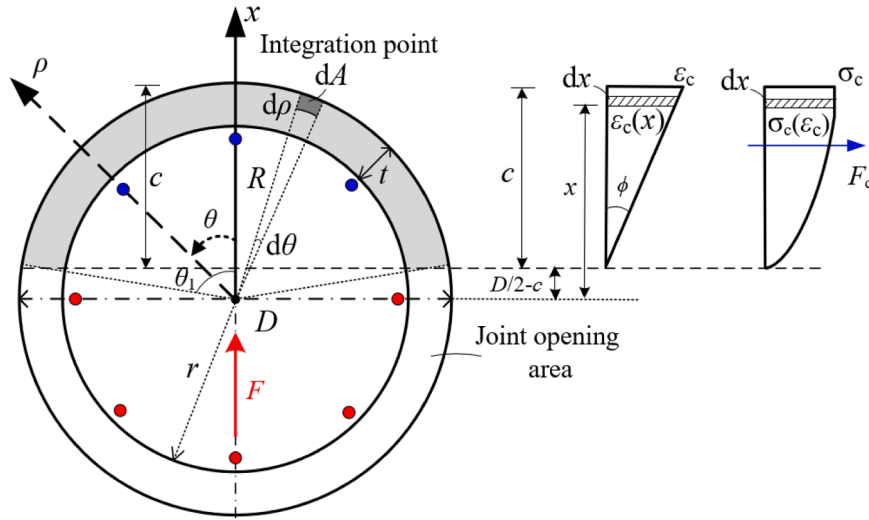


Fig. 24. Stress and strain analysis of the bottom section.

$$M = \int_A \sigma_c \cdot x dA + \sum_{i=1} \Delta M_{pt} \quad (33)$$

$$= \int_{-\theta_1}^{\theta_1} \int_r^R \sigma_c ((\rho \cos \theta - R + c) \cdot \phi) \cdot \rho \cos \theta \cdot \rho d\rho d\theta + \sum_{i=1} \Delta P_i \cdot (d_{p,i} - R)$$

The bending moment M is composed of lateral force F at the top of the tower and the second-order effect of the vertical load N , thus the lateral force F can be calculated as:

$$F = \frac{M - N \cdot \Delta}{H} \quad (34)$$

The second-order effect can be ignored if the prestress is significantly greater than the vertical load.

5.5. Prediction of load-displacement curve and ultimate bending capacity

Through the above equations, the lateral force F and horizontal displacement Δ at the top of the tower corresponding to any height of the compression zone c can be obtained. However, the curvature ϕ was unknown and must be calculated by Eq. (25). In Eq. (25), the compressive strain ϕ_c at the edge of the compression zone was also unknown. Consequently, it is necessary to iteratively update the curvature ϕ and its corresponding compressive strain ϕ_c , until the equilibrium condition in Eq. (31) is satisfied. The ultimate state of the tower is defined as the state when the compressive strain at the edge of the compression zone reaches the ultimate compressive strain ϵ_{cu} . Follows are the specific calculation steps:

- (1) Determine a height of compression zone c and select an initial curvature ϕ .
- (2) Calculate the rotation of joint θ_r .
- (3) Calculate the compressive strain at various positions within the height of the compression zone, and integrate to obtain the compressive force of concrete F_c by combining the stress-strain curve of concrete.
- (4) Calculated the variations of prestressing force through Eq. (32).
- (5) Check the equilibrium Eq. (31). Update the curvature ϕ and repeat steps (1) to (4) for iterative calculations until the equation is satisfied.
- (6) Calculate the bending moment M through the updated curvature ϕ according to Eq. (33).
- (7) Calculate and record the lateral force F and horizontal displacement Δ corresponding to the bending moment M .

- (8) Continuously update the compression zone height c and repeat steps (1) to (7) until the compressive strain exceeds the ultimate compressive strain ϵ_{cu} . End the iteration and plot the load-displacement curve by the recorded results.

Based on the above theoretical model and calculation steps, a comprehensive solution process for predicting the compression-bending behavior of prestressed concrete towers has been developed, as shown in Fig. 25. To validate the accuracy and applicability of the proposed theoretical model and the solution process for predicting the compression-bending behavior of prestressed concrete towers, five specimens from the experimental study were analyzed. The load-displacement curve at the top of the tower was obtained and compared with the measured curves, as shown in Fig. 26.

It can be seen that the theoretical curves for specimens agree with the measured skeleton curves well, demonstrating similar trends in both stiffness and ultimate capacity. Except for specimens CB-0.44-c and CB-0.44-m, the ultimate displacement predicted by theoretical curves were generally lower compared to the measured ultimate displacement. The reason is that the theoretical calculation was terminated when the compressive strain of the edge of the compressive side reached the ultimate compressive strain. However, it could be seen from the test phenomenon that the section can undergo a certain degree of stress redistribution due to the confinement after the ultimate compressive strain was exceeded, allowing the tower to continue bearing load without immediate failure, especially in towers with lower prestress ratios. Consequently, the ultimate displacements given by the theoretical model were lower, but the theoretical model could still provide a good prediction for ultimate bending capacity and the response of load-displacement before failure. The comparison between the theoretical ultimate capacity F_{uc3} and the measured ultimate capacity F_{ue} of specimens is shown in Table 5. The average value of F_{uc3}/F_{ue} is 0.958, with a standard deviation (SD) of 0.0516. Compared with two existing engineering design methods, the average relative error of the proposed method was reduced by 44.4 % and 15.1 %, respectively. This indicates that the theoretical calculated results agree better with the experimental results. The confinement effect of reinforcements and the compressive constraint of the foundation on the tower were not considered in the proposed theoretical model. However, considering that the prestressed concrete tower is a thin-walled structure, the stress redistribution and plastic development are limited. Consequently, the tower tends to undergo brittle failure especially when the prestress ratios are large, making it unsuitable to overly account for effects of confinement. Therefore, the proposed theoretical model can conservatively predict

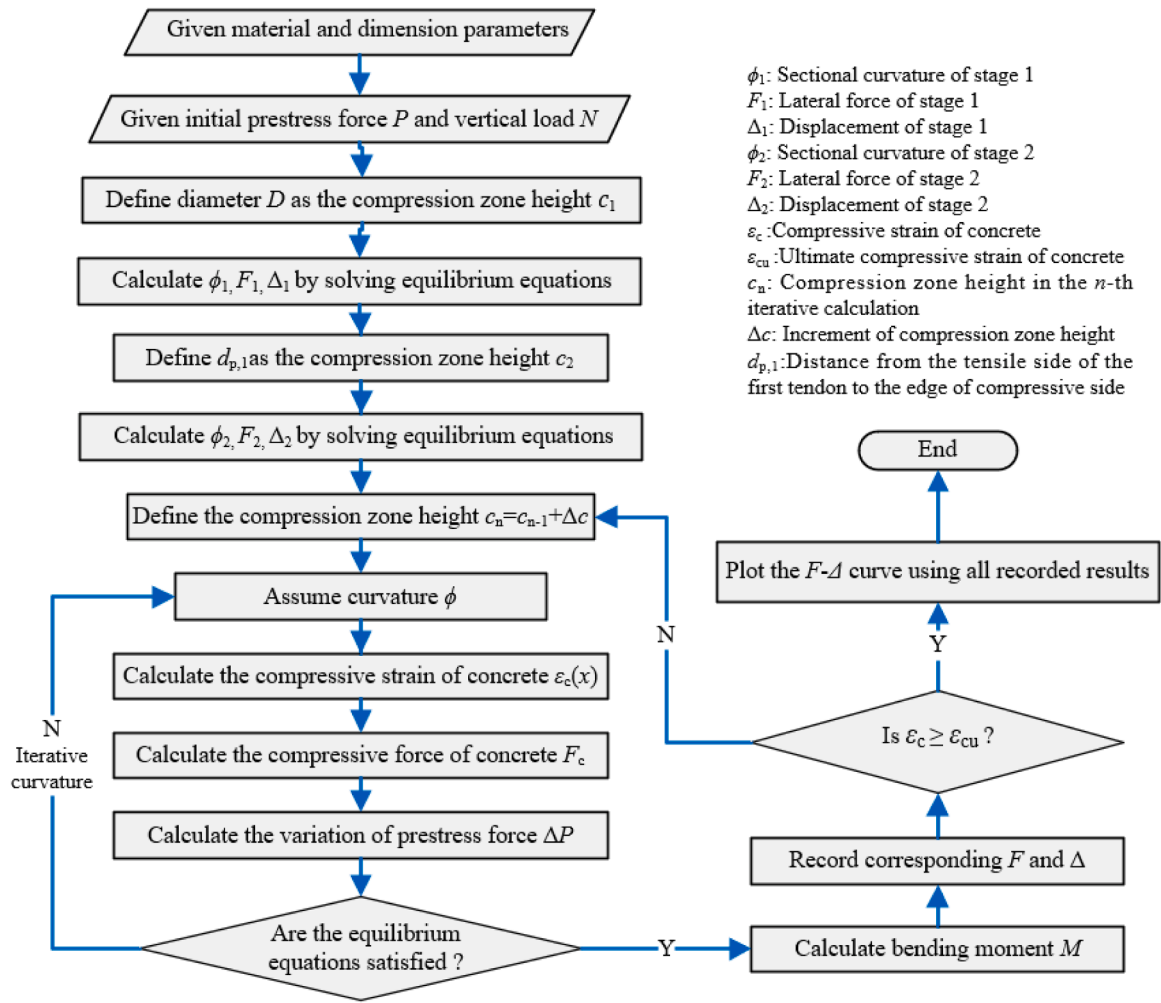


Fig. 25. Solution process of predicting the compression-bending behavior of prestressed concrete towers.

the ultimate capacity of prestressed concrete towers under compression-bending load, providing a theoretical reference for engineering design.

6. Conclusions

Experimental and theoretical studies on the compression-bending behavior of the thin-walled prestressed concrete tower for wind turbines are presented in this paper. The following conclusions can be drawn:

- (1) Five prestressed concrete tower specimens were tested under compression-bending loads, considering the effects of joint opening and prestress ratios. During the loading process, the horizontal joints exhibited repeated opening and closing. The typical failure mode of the specimens can be summarized as that the concrete in the bottom region exhibited significant compressive damage and subsequently crushed, while the horizontal joint opened. When the prestress ratio was high, the concrete experienced sudden large-area crushing, with the joints remaining more insignificantly open at failure. In contrast, in specimens with lower prestress, the concrete gradually spalled, and the joint opened more pronounced at failure.
- (2) The hysteretic curves exhibited significant pinching, with smaller enclosed areas. The compression-bending behavior is greatly affected by prestress ratios: a higher prestress ratio provided a larger sectional area at the bottom of the tower to resist bending moments, resulting in greater bending capacity. However, this

also led to reduced ductility and deformation capacity, causing rapid failure after reaching peak load. The ductility and load-bearing capacity of the monotonic loading specimen were only slightly higher than those of the cyclic loading specimen with the same prestress ratio.

- (3) As the prestress ratio of the specimens decreased, the ability of the tower to rotate around the joint increased, leading to a more gradual degradation of stiffness and a larger prestress increment. This also resulted in smaller energy dissipation and residual displacements before reaching peak load, and enhanced the self-centering performance. Conversely, specimens with higher prestress ratios exhibited smaller prestress increments, leading to significant energy dissipation and residual displacements after reaching peak load. These specimens showed greater irreversible damage and weaker self-centering performance, making their mechanical behavior closer to that of cast-in-place structures.
- (4) The accuracy and applicability of the existing engineering design methods for calculating the ultimate bending capacity were evaluated. In current calculation methods, the joints were assumed to not opened under bending moments, or the towers were directly compared to cast-in-place structures, making it impossible to accurately predict the bending capacity. But for towers with high safety redundancy requirements, the method based on plane-section assumption can still be used to conservatively check the bending capacity.
- (5) The full-process mechanism of prestressed concrete towers under compression-bending loads was analyzed. By combining the

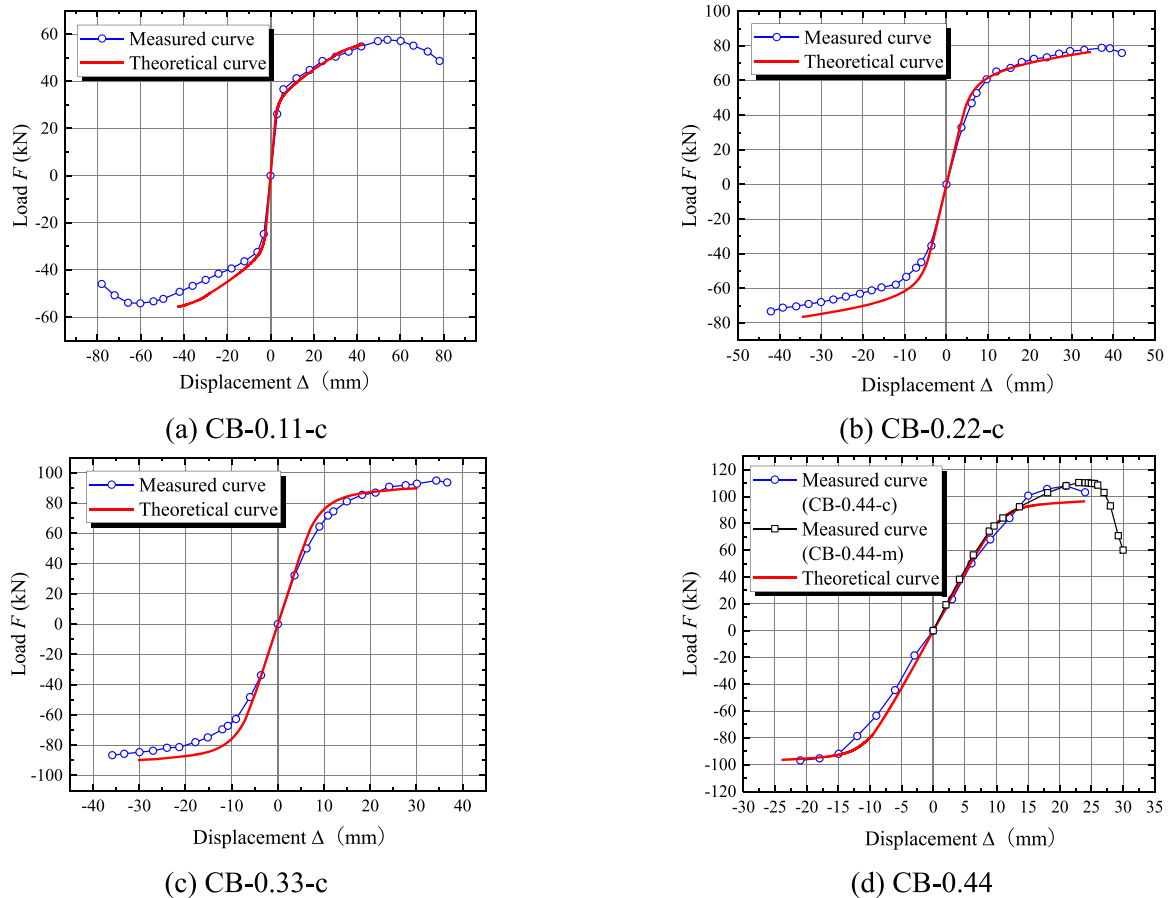


Fig. 26. Comparison of theoretical analysis and measured skeleton curves.

equivalent plastic hinge model and ultimate equilibrium theory, a theoretical model for the full-process analysis under compression-bending loads was established. An iterative solution process was developed based on the theoretical model to predict the load-displacement curves, and the theoretical calculation results agreed well with the measured results. The proposed theoretical model can be adopted to predict the compression-bending behavior of prestressed concrete towers and calculate their corresponding ultimate bending capacity.

CRediT authorship contribution statement

Wei Ren: Writing – review & editing, Visualization, Investigation, Formal analysis. **Xu-Hong Zhou:** Resources, Project administration, Conceptualization. **Ran Deng:** Writing – review & editing, Visualization, Validation. **Shou-Zhen Li:** Writing – review & editing, Validation. **Yu-Hang Wang:** Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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